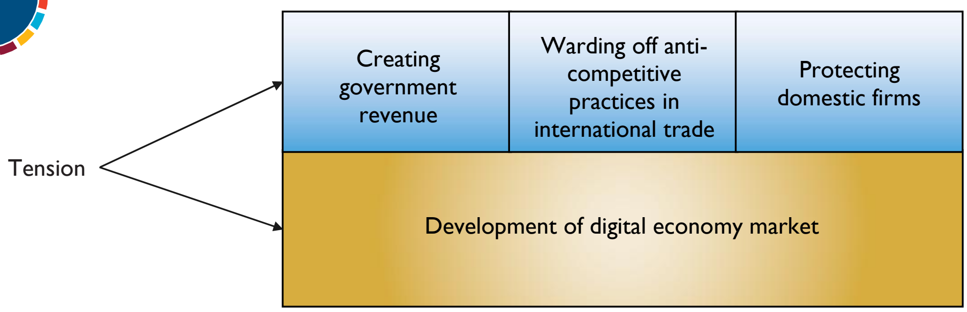
Pillar 1. Tariffs and trade defence

Pillar 1 covers tariffs and trade defence measures applied to intraregional imports of ICT goods. ICT products or ICT goods are final goods whose main purpose is to capture, transmit, process, and render information and intermediate goods and inputs that are crucial to manufacturing these goods. Examples include smartphones, computers, network equipment, storage media, semiconductors, electrical parts, electronics, sensors, processors and cables. The RDTII 2.1 results are based on ICT products under WTO Information Technology Agreement (ITA) I and ITA II, as well as the proposed ITA III expansion by the Information Technology and Innovation Foundation (ITIF) (see Annex III).[4]

As figure 5 reveals, these measures are often designed to protect domestic firms, provide a source of Government revenue, and counteract anti-competitive practices taken by foreign firms or foreign Governments. However, the measures have risks of limiting the development of the digital economy market. For example, tariff measures on electronics, telecommunication items and high-performance computing technologies may reduce an economy's exposure to advanced digital products or technologies, thereby deepening digital divides (Ferracane and others, 2019). Furthermore, tariffs and trade measures on basic materials for batteries and hardware, as well as finished products such as computers, electronics and telecommunication equipment, affect the costs of digital trade.

Figure 5 **Tension among different policy objectives in Pillar 1**



Pillar 1 measures consider the following areas:

- Effectively applied tariffs on ICT goods (in weighted average) imported from regional economies within the considered United Nations region;
- No duty-free tariff lines on ICT goods from other economies within the considered United Nations region;
- Not in the WTO Information Technology Agreement (ITA) of 1996 (ITA I) and its expansion in 2015 (ITA II); and
- Trade defence measures, including anti-dumping, countervailing duties and safeguards on ICT-related goods imported by other economies within the considered United Nations region.

[4] The RDTII 2.1 results are based on the list of ICT products found in the "ITA 3.0" list proposed by the Information Technology and Innovation Foundation (ITIF) 2021 (Ezell and Dascoli, 2021). The ITA 3.0 includes all products under the WTO's Information Technology Agreement (ITA) I and ITA II products as well as additional products provided by the ITIF. The proposed WTO ITA III expansion includes next-generation ICT products, such as robots, 3D printers, drones, certain medical technologies, and unmanned aerial vehicles.

Effectively applied tariffs on ICT goods (in weighted average) imported from other economies within the considered United Nations region

This indicator is the weighted average of effectively applied tariffs (AHS)[5] of each reporting economy to the rest of the economies within the considered United Nations region (e.g., Viet Nam effectively applied tariffs to the rest of the ESCAP (Asia-Pacific) region). Effectively applied tariffs are defined by the World Integrated Trade Solution (WITS) as the lowest available tariffs.If a preferential tariff rate exists, it will be used as the effectively applied tariff. Otherwise,a most-favoured nation (MFN) applied tariff will be used. The reason why the AHS also incorporates the preferential tariff is that the index seeks to measure the actual level of tariffs applied to the ICT goods (the WTO ITA I, ITA II, and the proposed ITA III expansion, see Annex III) from regional partners considered. Using the MFN tariff alone would only provide a picture of the highest tariff rates imposed.

The reason why the index uses the weighted average rather than the simple average rates is that the simple average does not account for the relative importance of ICT goods in terms of their trade volumes. The weighted average accounts for this relative importance by weighing each tariff rate by the share of the trade volumes of each tariff line. This means that tariff ratesof ICT goods with higher trade shares get higher weights than tariff rates of ICT goods with lower trade shares.

To normalize the tariff rates into the score between zero and 1, the score calculation follows a linear function of f(x) = 0.1x, where x is the weighted average tariff rate on ICT goods. If the average tariff rate is within the range of zero and 10%, the score will be less than ‘1’, while any tariff rate higher than 10% will be scored at ‘1’. For example, an economy with an average tariff of 1% receives a score of ‘0.1’, while another economy with an average tariff of 10% or higher receives a score of ‘1’.

The data for average tariff rates are found in the WITS database. For the specific steps to find the data for zero-tariff lines, refer to Annex I.

No duty-free tariff lines on ICT goods imported from other economies within the considered United Nations region

The second indicator is the no duty-free tariff lines or coverage rate of zero-tariffs that apply to ICT goods (the WTO ITA I, ITA II, and the proposed ITA III expansion) imported from other economies within the considered region. The indicator follows a linear function, f(x) = -0.025x + 1.75, where x is the coverage rate calculated from the number of free tariff lines for ICT goods divided by the total number of tariff lines for ICT goods, multiplied by 100. However, when the coverage rate is 30% or below, the score will be truncated to ‘1.’ In contrast, when the duty-free coverage rate is greater than 70%, the score will be truncated to ‘0’. The data for zero-tariff lines in a given economy is found in the WITS database. For the specific steps to find the data for zero-tariff lines, refer to Annex I.

[5] For more information about the different types of tariffs, please visit the WITS website: https://wits.worldbank.org/wits/wits/witshelp/content/data_retrieval/p/intro/c2.types_of_tariffs.htm.

Not in the WTO Information Technology Agreement (ITA I or ITA II)

This indicator looks at whether economies are members of the WTO's ITA I ("the Ministerial Declaration on Trade in Information Technology Products") or ITA II ("the Ministerial Declaration on the Expansion of Trade in Information Technology Products"). The ITA I requires its members to "eliminate and bind customs duties at zero for all products specified in the Agreement."[6] In ITA II, the members agreed to expand the products covered by ITA I by eliminating tariffs on an additional list of 201 products.[7]

The ITA I and II are a close proxy for tariff measures that apply to ICT products. The WTO reports that, today, ICT products account for approximately 10% of global merchandise exports (WTO, 2001). ICT products covered by the ITA I alone account for approximately 97% of world trade in the IT sector. The expanded list of the products under ITA II accounts for approximately 7% of total global trade today.

The reason why the index includes this indicator, even though it already takes into account the coverage rate of zero tariffs on ICT products, is that the schedules of concessions bind the economies to their obligations to other members under the agreements, unlike domestic policies of zero tariffs. Because the members would have the legal obligation not to impose import duties on the covered products, investors and traders would benefit significantly from improved market access, predictability and certainty (WTO, 2001).

The score is '0' if an economy has signed both agreements. If an economy signed only ITA I, the score is '0.5'. If an economy signed neither of them, the score is '1'.

To see participants in the Agreement, check the WTO official site. Specifically, to check whether the economies have ITA II membership, the official government website, the WTO Ministerial Declaration on the Expansion of Trade in Information Technology Products 2015, and the information provided on the WTO official site are useful sources. The secondary source should only serve to guide researchersto the primary sources.

Trade defence measures, including anti-dumping, countervailing duties and safeguards on ICT goods and ICT-related goods imported by other economies within the considered United Nations region

The indicator looks at whether an economy is enforcing trade defence measures such as anti-dumping duties, countervailing duties, and safeguard measures against ICT goods and ICT-related goods imported from any economy in the considered region.

Specifically, the 'ICT-related goods' refers to finished products and components used in the ICT equipment manufacturing, such as electrical connection terminals, aluminum alloy strips, power transformers, galvanized steel coils, stainless steel tubes, and glass fiber materials. These items serve as components or inputs in procuring finished products, including ICT goods in some cases. Raw materials used in ICT manufacturing, such as rare earth elements, lithium, cobalt, and silicon, are not considered under this definition.

Only the active measures will be counted, while the efforts of investigation measures in the pre-implementation stage or terminated measures will not be counted. For example:

[6] The ITA concessions are included in the participants' WTO schedules of concessions, and the tariff elimination is implemented on an MFN basis (WTO, 2001). This means that even economies that have not joined the ITA can benefit from the trade opportunities generated by ITA tariff elimination.
[7] For the list of products covered under the WTO's ITA I and ITA II, see Annex III.

- Argentina imposes an anti-dumping duty on imports of electrical connection terminals from China and Germany;
- Brazil imposes a definitive anti-dumping duty on imports of loudspeakers from China;
- India imposes an anti-dumping measure on imports of printed circuit boards originating in or exporting from China and Hong Kong (China), with duty rates between 30% and 76% depending on the exporting company;
- Türkiye imposes an anti-dumping duty on welded stainless-steel tubes, pipes and profiles originating in China, Malaysia, Viet Nam and Taiwan Province of China, with duty rates varying by the exporting company.

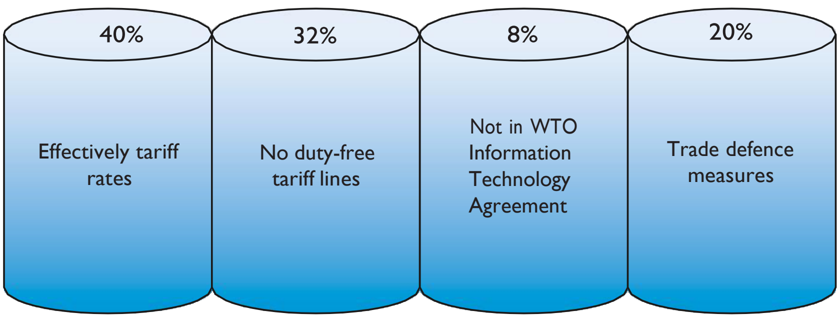
For each measure, the economy receives ‘0.25’, with ‘1’ being the maximum score in this indicator. Thus, if an economy is enforcing four or more of such measures, it receives ‘1’. The score is ‘0’ when a trade defence measure is not enforced.

For sources, find WTO members' anti-dumping notifications under the agreement on implementation of Article VI of GATT 1994 (“G/ADP/N/1/…”), notifications relating to countervailing measures or the most recent official gazette of a relevant national ministry. Useful secondary sources include the Integrated Trade Intelligence Portal (I- TIP) and the Global Trade Alert (GTA) database. The secondary sources should serve only to guide researchers’ attention to the primary sources.

The weights for each indicator

As shown in figure 6, weight rates of 40%, 32%, 8% and 20% are given to the indicators, respectively. The first and second indicators on tariffs applied to ICT goods are given great weight of 40% and 32%. This is because the tariffs (or lack thereof) on ICT products reflecta comprehensive impact on costs of digital trade in ICT products, whereas the trade defence measures sporadically apply to a few sets of imports, hence being given a lesser weight of 20%. Furthermore, the fact that an economy is not a member of the WTO ITA I or II does not mean that its tariff rates are restrictive, as long as the economy does not impose tariffs on a number of ICT products; thus, the last indicator was given the least weight of 8%.

Figure 6 **Pillar 1 indicators and the weights**



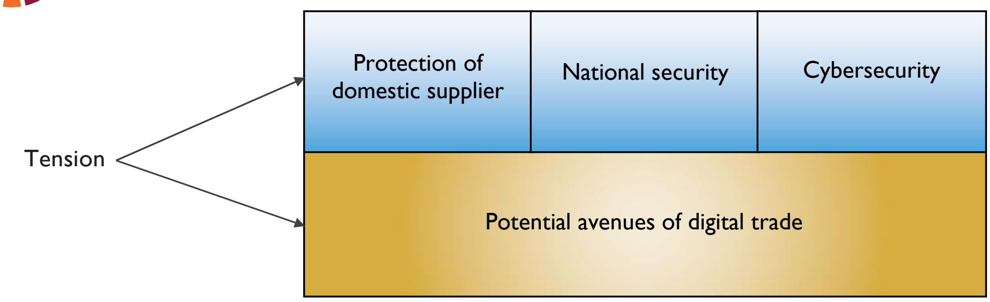
Pillar 2. Public procurement

Pillar 2 covers measures on public procurement related to ICT goods and online services. Public procurement is generally applied to various sectors, including those related to digital trade, affecting ICT goods and online services. There is tension among different policy objectives in procurement policies, as shown in figure 7. It investigates whether domestic bidders tend to have an advantage in Government procurement. Specifically, for procurement in sectors relevant to digital trade, such as IT technologies and infrastructure, national security and cybersecurity interests may be involved in forming procurement policies as procuring entities carry sensitive information in the operation of public administration.

However, some measures that exclude foreign suppliers in Government procurement block potential avenues of digital trade. Accordingly, while the national security and cybersecurity interests are significant, economies need to examine whether the measures are necessary to serve those interests or if other, less restrictive means exist.

Figure 7

Tension among different policy objectives in Pillar 2



Based on this understanding, Pillar 2 covers discriminatory measures or measures with high compliance costs. Specifically, the Pillar does so by looking at:

- Foreign exclusions from public procurement;
- Specific requirements on source codes, encryption and trade secrets;
- Limitations in procurement bidding; and
- Not in the WTO Government Procurement Agreement (GPA).

For this Pillar, useful secondary sources include the WTO Trade Policy Reviews Report, the OECD Services Trade Restrictiveness Index (STRI), and the World Bank-WTO Services Trade Restrictions Index (STRI). The secondary sources should only serve to guide researchers to the primary sources(i.e., actual laws, regulations and official documents issued by Governments).

Foreign exclusions from public procurement

This indicator covers measures that exclude foreign enterprises from public procurement, including for ICT goods and online services. The exclusion of foreign firms in public procurement not only discriminates against foreign firms but also limits the opportunity of the domesticeconomy to access new digital technologies that foreign firms could offer.

- The exclusion is when foreign enterprises are not allowed to participate in the public procurement of certainICT goods and online services, or when a list of specific economies, groups or companies are banned from participating in public tenders. For example, the Russian Federation has established a ban on the use of foreign messenger services in public procurement. Moreover, in Nigeria, Government entities must purchase hardware locally and procure software from local developers;
- Foreign enterprises can only take part in public tenders when domestic service suppliers are not participating or available. For example, Nepal accepts foreign suppliers only when competitive prices are not available from more than one local supplier, no national bids are submitted, or the goods are certified by a public entity as complex or specialized. Bolivia directs public calls for national purchases of up to approximately US$ 1,166,000 exclusively to legally established national production companies, with foreign companies allowed only if no national providers are available. Foreign enterprises can participate in public procurement when domestic enterprisesare required to bid in cooperation with foreign enterprises. For example, Thailand requires foreign consultants to team up with domestic ones to participate in public tenders.

A country receives a score of ‘1’ if it excludes foreign enterprises from public procurement. The score is also ‘1’ if there is an instance where an economy has excluded two or more specific (groups of) foreign firms from public procurement. The score of ‘0.5’ is assigned when there is an instance where an economy has excluded a specific (group) of foreign firm(s) from public procurement. The score is ‘0’ if no such measure exists. The score of ‘0’ also applies when there is only a legal basis that grants the Government the power to exclude foreign enterprises, because this does not automatically constitute an exclusion.

Specific requirements on source code, encryption and trade secrets

The indicator asks: (a) whether firms are required to surrender source code, encryption or other trade secrets, or to transfer ownership of exclusive rights to use IP such as patented technology, as a condition for successful public procurement and (b) whether firms are required to use a specific encryption standard to be successful for their bidding.

A requirement to transfer technology or use a specific encryption standard in public procurement often stems from interest in protecting national security. However, these types of requirements may prevent foreign companies from entering the domestic market because of concerns about the disclosure of their trade secrets and the increasing costs of configuration with a new system. For example:

- Egypt mandates the use of encryption requires the approval of the National Telecom Regulatory Authority as well as the armed forces and national security entities;
- Indonesia requires that providers of custom software for public electronic systems must submit source code and documentation associated with their service to the relevant public institution; and

- Kazakhstan requires that software providers submit source codes and configurations for inclusion in a unified Government register for public procurement.

The score is ‘1’ if there is a requirement to surrender such trade secrets as a condition for participating in tenders. The score is ‘0.5’ if firms are required to use specific encryption to win tenders. The score is ‘0’ if there is no such measure.

For this particular indicator, the World Map of Encryption Laws and Policies is a useful secondary source. The secondary sources should only serve to guide researchers to the primary sources.

Limitations in procurement bidding

This indicator covers limitations on participation in public procurement. These limitations can take various forms: (a) allocation of a quota; (b) preference in favour of certain suppliers; and (c) price preference in favour of certain suppliers.These limitations trigger when certain conditions are met, which include: (a) the nationality of suppliers; (b) other status such as being SMEs or indigenous; and (c) percentage of local content in supplies.For example:

- Argentina establishes that in public-private partnership contracts, at least 33% of the goods and services used must be provided by local companies;
- Azerbaijan grants a procurement discount of up to 20% in favour of locally produced goods;
- India reserves 25% of the annual procurement quota for micro and small enterprises, including a 3% reservation for women-owned micro and small enterprises;
- Indonesia requires a minimum 40% domestic content for transmission towers and steel-reinforced conductors in public procurement ;
- Rwanda imposes local content requirements giving a 15% preference to goods produced locally and a 10% preference to bidders registered in Rwanda; and
- Sri Lanka provides a 20% domestic preference margin for Government-funded procurements of manufactured goods that meet a 30% local-value additional test and other eligibility criteria, such as registration under the Companies Act and submission of an affidavit and audited financial statements.

These limitations discourage international procurement in sectors relevant to digital trade. To categorize a measure accordingly, the first step is to identify whether it constitutes a limitation rather than an exclusion mentioned in indicator 2.1, and the second step is to identify whether the measure is applied in a discriminatory manner against foreign suppliers or equally applied to all suppliers.

The score is ‘1’ for a measure that directly discriminates against foreign bidders, or the absence of institutional transparency, or if multiple non-discriminatory limitations (i.e., two or more measures falling under the ‘0.5’ category). Institutional transparency implies that all public procurement laws, regulations, guidelines, and bidding procedures are publicly available, easily accessible (e.g., on a government website), and clearly communicate the requirements and decision-making processes. The score is ‘0.5’ if there is a measure applied to all bidders, whether domestic or foreign, such as local content requirements and performance-based conditions. The score is ‘0’ if none of the above measures exist. The score is also ‘0’ when there is only a legal basis that grants the Government the power to impose limitations in procurement bidding, because this does not automatically constitute a limitation.

Not in the WTO Agreement on Government Procurement (GPA)

This indicator looks at (a) whether an economy is a signatory to the WTO's GPA and (b) whether its coverage schedule includes services relevant to digital trade (box 2). The GPA is a binding plurilateral agreement that promotes non-discrimination in public procurement by committing its members to the most-favoured-nation treatment and national treatment. However, these rules are subject to eachparty's coverage schedule (Annex V regarding services of the GPA)[8] that determines whether procurement activities in a particular sector are covered by the Agreement or not.

The score is '1' if an economy is not a member of the GPA or if an economy's coverage schedule does not cover any of the telecommunication services (CPC 752), telecommunications-related services (CPC 754), and computer and related services (CPC 84). The score is '0.5' if an economy is a GPA member and covers at least one but not all of these service sectors related to digital trade. The score is '0' if an economy,a member of the GPA, fully covers these service sectors related to digital trade (United Nations, 1991).

Each party's coverage schedule can be found on the e-GPA Gateway and the WTO GPA website.

[8] Available at: https://www.wto.org/english/tratop_e/gproc_e/gp_app_agree_e.htm

Box 2

The Agreement on Government Procurement (GPA)

The GPA is a plurilateral agreement within the framework of the WTO, meaning that not all WTO members are parties to the Agreement. As of July 2025, the Agreement has 22 parties comprising 49 WTO members (counting the European Union and its 27 member States as one party). Not long after the implementation of the GPA in 1994, the GPA parties initiated the renegotiation of the Agreement. The negotiation was formally adopted and came into force for all those parties to the GPA 1994 that had ratified the GPA 2012, while allowing other parties to the GPA 1994 to continue completing their domestic ratification procedures. The last of those other parties, Switzerland, completed the ratification in 2020, and the GPA 2012 replaced the GPA 1994.

The ultimate aim of the Agreement is to mutually open Government procurement markets among its parties by progressively reducing and eliminating discriminatory measures and achieving the greatest possible expansion of the coverage. According to the WTO, the GPA parties have opened procurement activities estimated to be worth more than US$ 1.7 trillion annually to international competition.

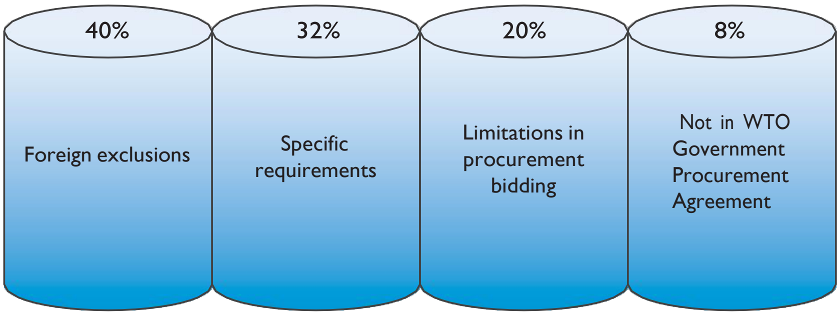
The GPA is composed mainly of the text of the Agreement and the parties' market access schedules of commitments. The text of the Agreement establishes rules mandating that open, fair and transparent conditions of competition be ensured in Government procurement in member economies. However, these rules do not automatically apply to all procurement activities of each party; rather, the coverage schedules of each party determine whether a procurement activity is covered by the Agreement or not. Only those procurement activities that are carried out by covered entities purchasing listed goods, services or construction services of a value exceeding specified threshold values are covered by the Agreement.

The weights for each indicator

As shown in figure 8, weights for each indicator are 40%, 32%, 20% and 8%, respectively. The exclusion of foreign firms from procurement receives the most significant weight of 40% because it discriminates against foreign firms and, *per se,* bans foreign participation in public procurement. The requirements regarding trade secrets and other limitations on participating in procurement receive less weight because they do not discriminate against foreign firms. Specific requirements to surrender source code, encryption and trade secrets are assigned a greater weight (32%) than the limitations on participating in procurement (20%). Although both indicators discourage a company from participating in public procurement, the specific requirements, including mandatory disclosure of trade secrets, affect the business's economic value and competition. The WTO GPA requires additional commitments of the most-favoured nation treatment and national treatment for procurement. However, not being a signatory of the treaty with a coverage schedule that includes services relevant to digital trade does not necessarily mean that the economy has a measure that discriminates against foreign firms or adds regulatory compliance costs in procurement. Therefore, the weight of this indicator is less than the others (8%).

Figure 8

Pillar 2 indicators and the weights

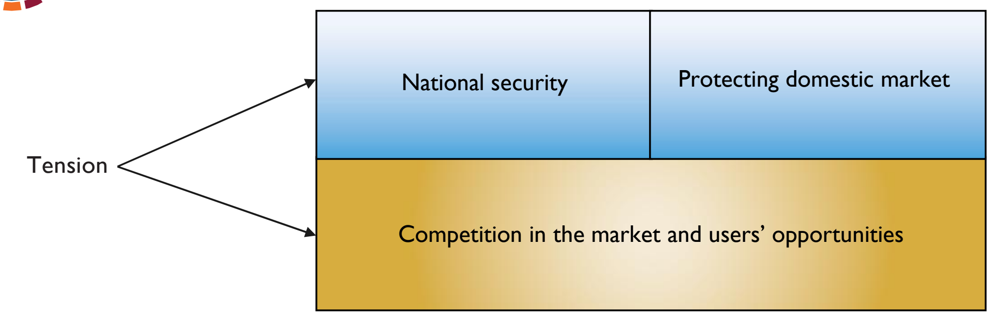


Pillar 3. Foreign direct investment

Pillar 3 covers measures on foreign direct investment (FDI) in sectors related to digital trade. These sectors include the manufacturing of telecommunication facilities, telecommunication services, computer services and Internet services.

Tensions in public and FDI policies may exist between the interests to protect national security and the domestic market, and the need to attract FDI (figure 9). Furthermore, as much as telecommunication facilities, networks and other related services are critical, economies may not want to increase their dependence on foreign investors in these sectors. However, some foreign investment measures do not necessarily address these concerns but may risk restricting foreign investment, reducing competition in the market, and limiting users' opportunities to access better quality goods and services.

Figure 9 **Tension among different policy objectives in Pillar 3**



In this regard, Pillar 3 covers conditions in foreign investment policy that may create a burden on foreign investors. Indicators in this Pillar consider the following requirements:

- Foreign equity limits;
- Joint venture requirements;
- Nationality or residency requirements for the board of directors or managers;
- Investment screening; and
- Commercial presence requirements.

For sources, these measures are found in laws governing companies, foreign investment or sectoral laws (e.g., telecommunications laws). Useful secondary sources include the WTO Trade Policy Reviews Report, the OECD STRI database, and the World Bank-WTO Services Trade Restrictions Index (STRI). The secondarysources should only serve to guide researchers to the primary sources.

Foreign equity limits

This indicator concerns the maximum foreign equity shares in sectors related to digital trade. Foreign equity shares are shares that foreign natural or legal persons hold in a firm incorporated in the investee economy. The indicator covers restrictions on foreign equity shares in (a) privately owned companies; and (b) Government-controlled companies—fully or partially state-owned enterprises (SOEs). Limitationon foreign equity shares is a direct obstacle for foreign investors that could induce a higher level of development in the sectors.

The foreign equity shares under this Pillar focus on those applied horizontally to all sectors or specific sectors relevant to digital trade, **except telecommunication**[9] **and e-commerce sectors, where they are captured separately under Pillars 5 and 12**. Certain economies regulate the telecommunication and e-commerce sectors through sector-specific regulators with jurisdiction over these areas. However, foreign equity caps applied under **a horizontal framework are recorded only under this indicator** to avoid double-counting.

The score is '1' if there is a ban on foreign ownership in at least one sector or if only a minority stake (less than 50%) is allowed in more than one sector. The score is '0.8' if only a minority stake is allowed in one relevant sector. The score is '0.5' where a controlling stake (more than 50%) is allowed, but maximum caps on foreign equity exist. The score of '0.5' also applies where limitations on foreign equity shares only exist in SOEs. The score is '0' if there is no limitation on foreign equity shares in relevant sectors.

Joint venture requirements

This indicator asks whether there is a requirement for firms to engage in a joint venture with a local firm in order to invest or operate in the economy. While forming a joint venture with a domestic firm helps foreign investors to strategize business effectively in a market unfamiliar to them and navigate through domestic regulation, the investors are in a better position than the Government to determine whether a joint venture is necessary. Furthermore, a mandatory requirement to form a joint venture with a domestic partner can discourage foreign investors due to the concern that technology might be forcefully transferred. Examples of joint venture requirements are:

- China, under a pilot policy, requires foreign providers of virtual private network (VPN) service to form a joint venture with Chinese partners, with foreign ownership capped at 50%. In addition, such operations must be conducted within designated pilot free trade zones;
- Egypt requires a joint venture for companies operating in trade sector projects, with possible exceptions for projects in remote areas;
- Liberia requires a joint venture or partnership between a Liberian anda foreigner to invest in a few businesses (commercial printing, advertising, graphics and commercial artists) if the total shareholding of the Liberian is at least 25% and the total capital invested is not less than US$ 300,000; and
- Vanuatu, requires foreign firms that undergo "expansion" more than three times to form a joint partnership with a citizen of Vanuatu.

If there is one of these measures, the score is '1.' Otherwise, the score is '0.

[9] Pillar 5 covers all forms of basic and value-added telecommunications services. Broadcasting services are not included under Pillar 5; any FDI measures, including foreign equity limits, are recoded under Pillar 3.

Nationality or residency requirements for the board of directors or managers in sectors relevant to digital trade

This indicator asks whether there are nationality or residency requirements for members of the board of directors or managers. These measures prevent foreign investors from appointing board members or managers of their choice.[10]

- Nationality requirements. For example, Bhutan requires that internet service provider (ISP) businesses must have local managers and personnel who are citizens of Bhutan. Indonesia reserves 19 positions, including directors, managers and supervisors for Indonesian nationals; and
- Residency requirements. For example, Palau, Kiribati and Singapore require that a company must have at least one director to be ordinarily resident in the economy. Georgia mandates that at least one manager of a limited liability corporation must have domicile within the economy.

If there is any requirement that at least one member of the board of directors or a manager has to reside in the economy or be a national of the economy, the score is ‘1’. Otherwise, the score is ‘0’.

Investment screenings

This indicator asks whether (a) an economy has adopted any screening mechanism for foreign investment or mergers and acquisitions (M&A) in sectors relevant to digital trade, excluding those implemented solely for antitrust purposes, unless such mechanisms are found to be discriminatory; and (b) whether there was a case of blocking investment or M&A based on the use of the concerned screening measure in sectors relevant to digital trade.

Investment screening refers to regulatory mechanisms that allow Governments to review, approve, or block foreign investments or acquisitions. The mechanisms are primarily based on either the interests of national security and public order or purely economic interests. Even the screening mechanisms that refer to economic interest, conditions and criteria are often unclear. The process mostly applies in a discretionary manner, creating uncertainty for investors and potentially discouraging investment activities. It is challenging to declare that screening mechanisms based on national security interests are less justified than those based on economic interests or vice versa.

Examples of the screening mechanisms in sectors relevant to the digital trade are:

- Armenia grants the authority to review foreign investments on the grounds of national security. The terms “national defence” and “security” are not clearly defined in the law.
- Kiribati requires foreign investors to obtain prior approval and a valid investment certificate before investing. The investment threshold of US$ 250,000 requires Cabinet approval.

[10] Residency or nationality requirements that apply only to officers who are not directors or managers do not count under this sub-pillar.

- In Mexico, the National Commission of Foreign Investment evaluates whether foreign investment applications meet certain criteria, including the impact on employment and workers' training, technological contribution and an increase in the competitiveness of the economy. This Commission can prevent acquisitions by foreign investment for reasons of national security;
- New Zealand applies a three-stage investment screening process. First, a foreign investor in a significant business asset (an investment activity that results in 25% or greater ownership or whose value exceeds US$ 71 million) must obtain consent. Second, a foreign investment in strategically important businesses, such as telecommunication infrastructure and media entities, can be subjected to the national interest test, and the consent may be declined if a transaction is contrary to national interest. Third, investment is subject to a 'call in transaction' under which the Government may block, impose conditions on, or order disposal of the activities if they pose a threat to national security or public order;
- Uganda's Investment Code Act stipulates certain screening measures for local and foreign investors intending to invest in information technology; and
- Vanuatu conditions its approval of investment activities on the provision of employment to locals and local capacity-building.

If there is a case where the screening mechanism has been used to block an investment in sectors relevant to digital trade, the score is '1'. If two or more mechanisms exist, the score is capped at '0.5'. For any screening mechanism, the score is '0.25'. If there is no screening mechanism, the score is '0'. Note that anti-trust measures related to M&A are not considered a restriction, unless discriminatory.

Commercial presence requirements

This indicator asks whether an economy requires foreign service providers to establish a commercial presence. This refers to services provided within an economy by a locally established office, branch, subsidiary or representative office of a foreign-owned and controlled company to offer cross-border services in sectors relevant to digital trade, corresponding to Mode 3 under the WTO GATS.[11] However, it does not cover requirements limited to business registration that do **not** involve the establishment of a physical or operational presence, for example:

- Colombia requires foreign companies to set up a branch in the country to engage in permanent business. These include activities such as opening business offices and intervening as a contractor of works or in the provision of services, among others;
- Lao PDR requires companies operating a satellite communication business to establish a registered business with a local office;
- Malaysia requires that a foreign company that carries out business in Malaysia to incorporate with a local company or register a branch within the economy;
- Pakistan requires significant social media companies—those with over 500,000 users or listed by the Authority to establish a permanent registered office in the country, preferably in Islamabad;

[11] For more information, see definition of services trade and mode of supply available at https://www.wto.org/english/tratop_e/serv_e/cbt_course_e/c1s3p1_e.htm.

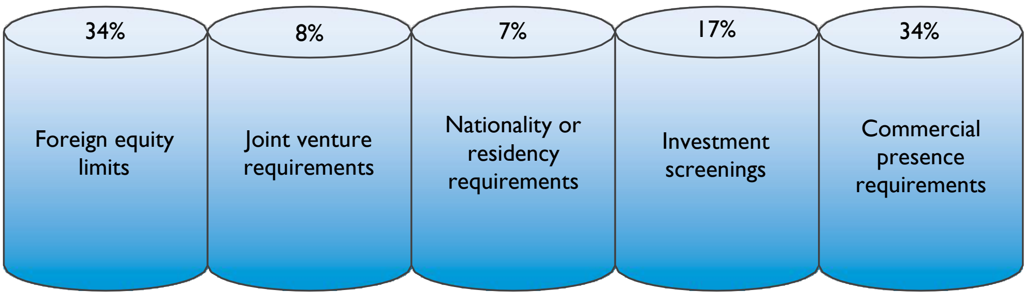
- Kyrgyzstan requires entities seeking a telecommunications license to establish a local presence by submitting legal registration documents, taxpayer identification, and other local documentation as part of the licensing process; and
- Viet Nam requires businesses providing specific categories of online gaming services to local users to establish a company in Viet Nam in order to obtain a license to operate.

For scoring, if at least one commercial presence requirement is met, the score is ‘1’. Otherwise, the scoreis ‘0’.

The weights for each indicator

As shown in figure 10, the weights are 34%, 8%, 7%, 17% and 34%, respectively. The greatest weight, 34%, is given to the first and the fifth indicators – foreign equity limits and commercial presence requirements. Numerical limitations on foreign ownership flatly discriminate against foreign investors and directly block foreign investment. The commercial presence requirements imply that digital trade under Service Trade mode 1 is not feasible. In addition, the requirements incur significant costs of infrastructure establishment and human resources in the recipient country. The fourth indicator – screenings of investment and acquisitions – is given the weight of 17% because they discourage foreign investors due to the time and cost involved in the process. The screening mechanisms could have less certainty and may block investments, thereby receiving higher weight than the joint venture requirements (8%) and nationality or residency requirements for directors or managers (7%). For the second indicator – a joint-venture requirement, which, while it discourages foreign investment, arguably due to the concern for technology transfer, it does not block foreign investment *per se,* as the maximum caps on ownership do. The third indicator – nationality or residency requirements for directors or managers
– is given the least weight since the degree to which these requirements discourage foreign investment is relatively limited.

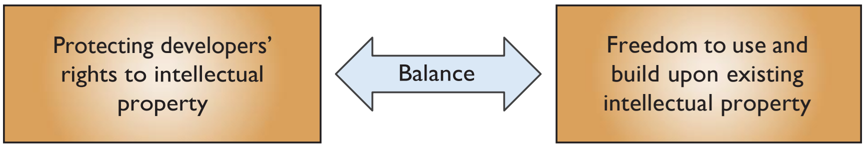
Figure 10 **Pillar 3 indicators and the weights**



Pillar 4. Intellectual property rights

Pillar 4 deals with intellectual property rights (IPRs), which are patents, copyrights and trade secrets. Since sectors relevant to digital trade are knowledge-intensive, IPRs play a crucial role in fostering innovation and creativity in digital trade. Generally, as shown in figure 11, sound policies regarding IPRs find a proper balance between the interests of protecting individual rights to intellectual property and fostering the freedom to use and build upon existing intellectual property. On the other hand, IP policies based on an ill-conceived balance tend to restrict digital trade because they contribute to creating an uncertain regulatory environment.

Figure 11 **Balance in different policy objectives in Pillar 4**



Pillar 4 comprises the following indicators:

- Patent application issues;
- Patent enforcement issues: civil and administrative procedures, and provisional measures;
- Patent enforcement issues: others;
- Not in the WIPO Patent Cooperation Treaty (PCT);
- Lack of copyright framework and exceptions;
- Online copyright enforcement issues: civil and administrative procedures, and provisional measures;
- Not in the WIPO Copyright Treaty (WCT);
- Not in the WIPO Performances and Phonograms Treaty (WPPT);
- Mandatory disclosure of trade secrets; and
- Lack of effective trade secrets legal framework.

Patent application issues

This indicator covers measures on the application process for patents, including differential treatment between local and foreign applications, and the measures applied to both local and foreign applications. The indicator asks whether (a) there is a restriction on the patent application process, and (b) what type of restriction. There are various forms of restrictions, such as a rejection of the patent application in a discretionary manner, a requirement to file a patent locally before filing abroad, or a requirement to translate documents into the official languages.

For example:

- A requirement to appoint a local agent. For example, Bhutan mandates that applicants residing or having their principal place of business outside Bhutan must be represented by a legal practitioner residing and practicing in the economy or a registered industrial property agent;

- A requirement to translate the application into a local language. For example, Kyrgyzstan requires patent applications to be submitted in the state or official language, that is, Kyrgyz or Russian. Foreign applicants may file the application in another language, provided that they include a translation into one of these two languages;
- A requirement to satisfy local requirements before filing abroad. For example, China requires that, where a Chinese entity or individual intends to file a patent application in a foreign economy for an invention made in China, the applicant must submit a request for a confidential examination to the patent administration department.This provision is applicable to all Chinese entities, including a subsidiary of a foreign company which is also considered a Chinese entity; and
- A requirement to register foreign patents locally. For example, Nepal introduces a regime where foreign patents are not valid in Nepal unless they are registered in Nepal.

For scoring, the score '1' is assigned when there exists a requirement to appoint a local representative, or a rejection of a patent application in a discriminatory manner. The score of '0.5' is assigned when there are procedural requirements, such as the obligation to file a patent locally before filing abroad. The score is '0' if there is no restriction in the patent application process.

Patent enforcement issues: civil and administrative procedures, and provisional measures

This indicator asks whether an economy's legal framework concerning patents incorporates (a) civil and administrative procedures; and (b) provisional measures.[12]

According to the WTO Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement, Part III enforcement of intellectual property rights, Articles 42-49 (Section 2 civil and administrative procedures and remedies),[13] right-holders are permitted to pursue civil and administrative procedures in cases of intellectual property rights infringements. Judicial authorities are empowered to request the infringers to pay the right holder damages adequately to compensate for the arising injuries.

Moreover, regarding the provisional measures, the WTO TRIPS Article 50 (Section 3 provisional measures) prescribes that the judicial authorities shall have the authority to order prompt and effective provisional measures: (a) to prevent an infringement of any intellectual property rights from occurring, and in particular to prevent the entry into the channels of commerce in their jurisdiction of goods, including imported goods immediately after customs clearance; and (b) to preserve relevant evidence in regard to the alleged infringement. If the provisional measures are implemented, the parties involved shall be given notice without delay. The provisional measures often include temporary remedies, such as the seizure or detention of suspected infringing products, or orders to preserve documents and other materials.

The legal framework considered under this indicator is not limited to dedicated intellectual property or patent laws. General laws, such as the Civil Code or procedural laws, are applicable. For example:

- In Azerbaijan, patent infringement is addressed through administrative, civil and criminal procedures. If parties cannot agree on compensation, disputes are resolved in court under the Civil Procedure Code. Provisional measures are authorized by the courts under the Civil

[12] **Civil Procedures** is a formal lawsuit between private parties where remedies like compensation for losses and court orders to stop the infringement) are sought. **Administrative Procedures:** This involves using the processes of a **government agency** (like a patent office or a specialized intellectual property enforcement body) to address patent infringement. **Provisional measures** are urgent, temporary legal orders, usually from a court, to immediately prevent further patent infringement or preserve crucial evidence while a full legal case is still ongoing.

[13] See https://www.wto.org/english/docs_e/legal_e/27-trips_04d_e.htm

Procedure Code and the Law on Enforcement of Intellectual Property Rights and Combating Piracy, which also governs enforcement and remedies; and

- In the Philippines, patent holders can seek civil remedies such as damages, attorney's fees, injunctions, and destruction of infringing goods, with provisional measures like temporary injunctions available under the Intellectual Property Code. Severe cases may lead to triple damages and criminal penalties, including imprisonment and fines.

For scoring, if an economy's legal framework concerning patents contains a full mechanism: civil and administrative procedures, and provisional measures, the score will be '0'. If an economy adopts either civil/administrative procedures or provisional measures, the score will be '0.5.' In the absence of these components, the score of '1' will be assigned.

Patent enforcement issues: others

This indicator covers measures on the enforcement of patents. The indicator asks whether there is a restriction on patent enforcement and the scope of such restriction.

The measures that will get scored may include, for example, different terms of patent protection that discriminate against the enforcement of foreign patent applicants or when patent holders' rights are restricted by state intervention.[14] Whether patent enforcement restrictions are applied horizontally across all sectors or limitedly to specific circumstances or sectors can have different impacts on businesses.

Exceptions to patent rights allowed under the WTO TRIPS, notably Articles 30 (exceptions to rights conferred), 31 (other use without authorization of the right holder) and 31bis, are not considered as restrictions on patent enforcement issues. Specifically, Article 30 allows limited exceptions to patent rights as long as they do not unreasonably conflict with the normal exploitation of the patent or prejudice the legitimate interests of the patent owner.[15] Article 31 and Article 31bis are about 'compulsory license, permitting the Government use to address public health, national emergencies, or other circumstances of extreme urgency, with adequate remuneration to the patent holder.[16] These exceptions are in place to balance patent rights and public interest, rather than constituting enforcement restrictions.

For scoring, the score is '1' when a restriction on patent enforcement is considered to have a high impact. This occurs, for example, when the issue is pervasive, affecting all circumstances and sectors, or if there are two or more patent enforcement measures applied to a limited case or a specific sector. The score is '0.5' if a measure is applied to a limited case or a specific sector or is considered as having a low impact. An example of a low-impact measure is a requirement to appoint a local agent for patent-related matters. The score is '0' if no such measure exists.

[14] Different terms of patent protection that apply to foreign applicants than to domestic applicants would likely mean that the economy does not accord national treatment because it is not a party to the Paris Convention or the WTO, which is very rare, or the economy imposes this restriction as an exception to the Paris Convention or the TRIPS Agreement.
[15] See https://www.wto.org/english/res_e/publications_e/ai17_e/trips_art30_jur.pdf
[16]See https://www.wto.org/english/res_e/publications_e/ai17_e/trips_art31_oth.pdf and https://www.wto.org/english/res_e/publications_e/ai17_e/trips_art31_bis_oth.pdf

Not in the WIPO Patent Cooperation Treaty (PCT)

This indicator looks at whether an economy is a member of the WIPO Patent Cooperation Treaty (PCT) (box 3). The PCT is a multilateral treaty and creates a unified patent system, which provides several advantages to residents or nationals of PCT members. The lack of PCT membership can increase the risks of elevated regulatory compliance costs, especially when the businesses wish to establish a patent in other member economies.

The score is ‘1’ if an economy is not a member of the PCT. Otherwise, the score is ‘0’. To see contracting parties in the Treaty, check the PCT Applicant’s Guide (national phase) or the official government website.

Box 3

The Patent Cooperation Treaty (PCT)

The PCT was concluded in 1970, and as of 2025 had more than 158 Contracting States. The PCT facilitates patent protection for an invention simultaneously in many economies, consisting of “international” and “national” phases. It begins with the filing of a single “international” patent application. The granting of patents remains under the control of the national or regional patent offices in what is called the “national phase”.

The PCT has several advantages. First, an international patent application under the PCT provides the applicant with an international search report and a written opinion on the potential patentability of the patent in the member economies. Although the international patent application alone does not grant a patent unless an application is filed subsequently in a national or regional patent office of the territory where the applicant wishes to establish a patent (or “enters the national phase”): (a) the applicant can refer to these documents to assess the worthiness of filing a patent in national or regional patent offices of the members; and (b) the process of patent prosecution in the national phase becomes easier due to these documents. The applicant may also request a supplementary international search report and international preliminary examination, which is the second evaluation of patentability.

Second, under the PCT, applicants have additional time to decide whether to file in a national or regional patent office to get a patent without worrying that the same invention would get patented in the meantime. This is because the date of filing under the PCT becomes the priority date. The effect of the priority date is that a patent does not become invalidated by reason of any acts by interval, such as another filing, publication or sale of the invention. This time could be up to (a) 18 months after an applicant files an international patent application or (b) 30 months after an applicant files with a national or regional patent office of a member economy.

Lack of a copyright framework and exceptions

This indicator asks for the presence of a copyright legal framework and what type of copyright exceptions, if any, an economy has adopted. The exceptions allow lawful use of copyrighted works without obtaining permission or a license from the copyright holders. This encourages foreign people (natural or legal) to use existing materials copyrighted in an economy and thereby foster innovation and development. However, the degree to which the exceptions promote this interest differs depending on the type of exceptions.

First, the doctrine of fair use provides that if the use of a copyrighted work is fair, the use is lawful. Such use is considered fair in light of several factors, such as: (a) the purpose and characterof the use; (b) the nature of the copyrighted work; (c) the amount and substantiality of the portiontaken; and (d) the effect of the use upon the commercial market. Thus, the doctrine is a flexible, case-by-case test, creating more room for new, innovative use. In general, the use of copyrighted material for criticism, comment, news reporting, teaching, scholarship or research is fair, but not always; furthermore, the use of other than these examples can be fair.

Second, the doctrine of fair dealing provides that the use of copyrighted material is permissible only if the use (a) falls under an exhaustive list of permissible uses, and (b) is fair (e.g., giving proper attribution to the copyright holder). Generally, the list is confined to research, private study, education, satire, parody, criticism, review or news reporting, leaving little 'wriggle room' for subsequent use of copyrighted works.

Aside from fair use and fair dealing doctrines, an economy may incorporate a test similar to the three-step test and other forms of copyright exceptions. The three-step test was established under Article 9(2) of the Berne Convention for the Protection of Literary and Artistic Works, which states that: "reproduction [of literary and artistic works protected by the Convention] in certain special cases (a) does not conflict with a normal exploitation of the work and (b) does not unreasonably prejudice the legitimate interests of the author." However, this test alone does not say much about what constitutes permissible use, thus creating uncertainty.

Additionally, examples of other forms of copyright exceptions can be found in Indonesia and the Marshall Islands, where a closed list of exceptions has been adopted rather than a flexible case-by-case test:

- Indonesia narrowly defines exceptions which allow limited uses of copyrighted works, such as for educational purposes, research, personal use, public interest, or review, provided that the use is non-commercial; and
- The Marshall Islands prohibits unauthorized commercial reproduction of sound recordings and audiovisual works but provides specific exceptions. These include works not created in the Marshall Islands, works by non-citizen owners, reproduction for free radio or TV broadcasts, historic or archival purposes, or for strictly personal use without commercial intent.

The score is '1' if an economy lacks a copyright legal framework or has no copyright exceptions. The score '0.5' is assigned when an economy adopts copyright exceptions, but they are not an explicit fair use or fair dealing regime, such as the three-step test and other types of copyright exceptions. The score is '0' if an economy adopts clear copyright exceptions following a fair use and/or fair dealing regime.

Online copyright enforcement issues: civil and administrative procedures and provisional measures

This indicator asks whether an economy's intellectual property rights legal frameworks concerning copyright incorporate (a) civil and administrative procedures; and (b) provisional measures.

According to the WTO Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement Part III, enforcement of intellectual property rights, Articles 42-49 (Section 2 civil and administrative procedures and remedies),[17] right-holders are permitted to pursue civil and administrative procedures in cases of intellectual property rights infringements. Judicial authorities are empowered to request the infringers to pay the right-holder damages adequately to compensate for the arising injuries.

Moreover, regarding the provisional measures, the WTO TRIPS Article 50 (Section 3 provisional measures) prescribes that the judicial authorities shall have the authority to order prompt and effective provisional measures: (a) to prevent an infringement of any intellectual property rights from occurring, and in particular to prevent the entry into the channels of commerce in their jurisdiction of goods, including imported goods immediately after customs clearance; and (b) to preserve relevant evidence in regard to the alleged infringement. If the provisional measures are implemented, the parties involved shall be given notice without delay.

The legal framework considered under this indicator is not limited to dedicated intellectual property or patent laws. General laws, such as the Civil Code or procedural laws, are applicable. For scoring, if an economy's legal framework concerning patents contains a full mechanism: civil and administrative procedures, and provisional measures, the score will be '0'. If an economy adopts **either** civil/administrative procedures **or** provisional measures, the score will be '0.5'. In the absence of these components, the score of '1' will be assigned

Not in the WIPO Copyright Treaty

This indicator looks at whether an economy is a signatory to the WIPO Patent Cooperation Treaty (WCT) (box 4). The WCT is a special agreement under the Berne Convention for the Protection of Literary and Artistic Works and ensures the protection of works and the rights of their authors in the digital environment.

The WCT expands the scope of copyright protection to two subject matters for at least 50 years: (a) computer programs, whatever the mode or form of their expression; and (b) compilations of data ("database"), in any form, whose contents constitute intellectual creation.[18] The Treaty grants exclusive rights apart from the rights recognized by the Berne Convention to authors, which are the right of distribution, the right of rental and the right of communication by wire or wireless transmission.[19]

The score is '1' if an economy is not a member of the WCT. Otherwise, the score is '0'. To see contracting parties in the Treaty, check the WIPO official site or the official government website.

[17] See https://www.wto.org/english/docs_e/legal_e/27-trips_04d_e.htm
[18] If a database does not constitute such a creation, it is outside the scope of this Treaty. For more information, available at https://www.wipo.int/treaties/en/ip/wct/summary_wct.html
[19] For the rights granted to authors, (a) the right of distribution is the right to authorize the making the copyright works available to the public through sale or transfer of ownership, (b) the right of rental is the right to authorize commercial rental of the copyrightwork to the public, and (c) the right of communication to the public is the right to authorize any communication by wire or wireless means, including though the Internet.

Not in the WIPO Performances and Phonograms Treaty (WPPT)

This indicator looks at whether an economy is a signatory to the WIPO Performances and Phonograms Treaty (WPPT) (box 4). The WPPT ensures the rights connected to aural performances, which are the subject matter of phonograms.[20] Two kinds of beneficiaries in the digital environment are focused on: (a) performers (actors, singers, musicians, etc.) and (b) producers of phonograms (persons or legal entities that take the initiative and have the responsibility for the fixation of sounds). The term of protection of these rights is 50 years.

The Treaty grants exclusive rights to the performances fixed in phonograms (excluding audiovisual fixations, such as motion pictures) and phonograms. The rights are the right of reproduction, the right of distribution, the right of rental and the right of making available by wire or wireless transmission.[21] The performers of the unfixed (live) performances also receive the right of broadcasting, the right of communication to the public, the right of fixation, as well as the moral right that is the right to claim to be identified as the performer and object to any distortion or modification that would be prejudicial to their reputation.

The score is ‘1’ if an economy is not a member of the WPPT. Otherwise, the score is ‘0’. To see contracting parties in the Treaty, check the WIPO official site or the official government website.

[20] According to Article 2(b) of the Treaty, “Phonogram” means the fixation of the sounds of a performance or of other sources, or ofa representation of sounds other than in the form of a fixation incorporated in a cinematographic or audiovisual work. Also, under Article 2(c), “Fixation” means the embodiment of sounds, or of the representations thereof, from which they can be perceived, reproduced or communicated through a device.

[21] For the rights granted to authors, (a) the right of distribution is the right to authorize the making the copyright works available to the public through sale or transfer of ownership; (b) the right of rental is the right to authorize commercial rental of the copyrightwork to the public; and (c) the right of communication to the public is the right to authorize any communication by wire or wireless means, including though the Internet.

Box 4

The WIPO Copyright Treaty (WCT) and the WIPO Performances and Phonograms Treaty (WPPT)

The WCT and WPPT were known as the “Internet Treaties”. Both treaties were concluded in 1996 and entered into force in 2002. As of 2025, both treaties have more than 100 Contracting States. The purpose of the two treaties is to update the existing WIPO Treaties on copyright and related rights in accordance with the development of digital technologies and the dissemination of protected material over digital networks. Thereby, these treaties contain several provisions related to the digital agenda, namely the rights applicable for the storage and transmission of works in digital systems, the exceptions to rights in a digital environment (the three-step test under the Berne Convention), and the technological measures of protection and rights management information.

The contracting parties will adopt the Treaties in accordance with their legal systems and oblige insurance that their enforcement procedures are available to protect against copyright infringement. Although the exercise of rights may be difficult to apply sufficiently in the online environment or the digital uses of works, the ‘technological protection measure’ (TPM) and ‘the rights management information’ (RMI) have been introduced to address such concerns in both Treaties.

The **‘obligation concerning technological protection measures’** prescribes that the contracting parties shall provide legal protection and effective legal remedies against unauthorized circumvention of effective technological measures. The TPM protects against circumvention of technologies that control access to copyright works. The circumvention of technologies refers to decrypting an encrypted work, avoiding, bypassing, removing, deactivating, or impacting a technological measure without authorization by the copyright owner. The protection can be components, software or any devices, such as passwords or encryption keys, that are capable of protecting the copyright from being copied or accessed.

The **‘obligation concerning rights management information’** prescribes that the parties shall provide adequate and effective legal remedies to a person having reasonable grounds to know that they remove or alter electronic right management without authority; or distribute or make copies of the copyright works available by knowing that electronic rights management has been removed or altered without authority. The RMI protects electronic rights management information against authorized access. The information means the information that identifies the work, the author, the owner of any right or the terms and conditions, and any numbers or codes that represent such information when any of this information is attached to a copy of a work or appears in connection with the communication of a work to the public. This information is necessary for the management of the rights.

Mandatory disclosure of trade secrets

This indicator asks whether a Government mandates the disclosure of trade secrets, suchas source code and algorithms. The disclosure requirement under this Pillar does cover disclosure requirements related to public procurement or encryption, as these are captured separately under Pillars 2 and 11, respectively. For example:

- In Malawi, there are requirements for encryption service providers to declare the means of encryption and the source code of the software used by the Malawi Communications Regulatory Authority;
- In Solomon Islands, disclosure of trade secrets to panel members is mandatory during proceedings, but safeguards exist as members are prohibited from further disclosing such information; and
- In Turkmenistan, authorities have the right to access information constituting a trade secret within their legal competence. While disclosure is mandatory, the law prescribes that such information must be handled in accordance with the established procedures and cannot be further disclosed unless allowed by law.

In line with the spirit of WTO Trade-Related Aspects of Intellectual Property Rights (TRIPS) to guide trade secret protection, “members shall protect the submission of undisclosed data against unfair commercial use and disclosure, except where necessary to protect the public, or unlesssteps are taken to ensure that the data are protected against unfair commercial use”. Hence, the requirement could be of limited scope, meaning that the disclosure is conditional or becomes mandatory only in certain circumstances. For example, an economy may adopt an escrow requirement for source codes in public procurement. Suppliers transfer the source codes to an escrow, and the source codes would be transferred to the Government, for example, when the suppliers go bankrupt or refuse to fix their products or programmes. Indonesia has an escrow requirement for custom-made software.

For scoring, it will not get scored when the Government mandates to disclose trade secrets and protects such information against unfair commercial use. If the Government does not provide safeguards against unfair commercial use, a disclosure requirementof limited scope affecting only specific types of products or specific circumstances (for example, disclosure under the national security threat provision[22] for certain companies) will get a scoreof ‘0.5’. The score is ‘1’ if there is more than one such measure or if the requirement is affectingan entire sector or all sectors horizontally. If there is no such measure, the score is ‘0’.

[22] GATS Article XIV bis stipulates Security Exceptions, allowing members the flexibility to implement a measure which is considered necessary for the protection of their essential security interests. However, the absence of clarity in defining and determining national security provisions introduces potential obstacles to digital trade. Due to the challenge of interpreting the ambiguous nature of national security, a score of 0.5 or 1.0 is assigned when a country implements a disclosure requirement based on these objectives, depending on the scope of such measures. For more information about GATS Article XIV. https://www.wto.org/english/res_e/publications_e/ai17_e/gats_art14_bis_oth.pdf

Lack of effective trade secrets legal framework

This indicator asks whether an economy has adopted a trade secrets legal framework that provides effective protection of trade secrets. **'Effective protection'** means that confidential business information or commercially valuable data are protected from unauthorized acquisition, use or disclosure through clear and enforceable legal provisions or well-established common law principles. The framework should enable right holders to take legal action against misappropriation, including remedies such as injunctions, damages and, where applicable, criminal sanctions.

Specifically, in common law jurisdictions, effective protection may arise from case law and judicial precedents on confidentiality and breach of confidence, even if there is no dedicated statutory framework. Therefore, the absence of a standalone trade secrets legal framework does not necessarily imply a lack of effective protection, provided that courts enforce protections against unauthorized disclosure of trade secrets. For instance, Singapore's trade secrets protection is grounded in the common law doctrine of breach of confidence. The framework and case law[23] meet international standards under TRIPS Article 39,[24] providing protection against unauthorized acquisition, use or disclosure and implementing remedies.

The scope of the indicator is the protection of trade secrets under domestic laws, i.e., statutes (intellectual property law and other laws), case laws, and other relevant legal frameworks. All forms of trade secret legal framework, such as Acts, provisions, clauses, or common law doctrine, are taken into consideration.

For scoring, the score is '1' if an economy lacks a trade secrets legal framework that is able to provide effective protection. The score will be '0.5' if it takes the form of more limited in scope or only partially enforced, i.e., when trade secrets are covered under narrow provisions without comprehensive coverage. The score is '0' if there is a presence of a regime for effective trade secrets protection in any form.

The weights for each indicator

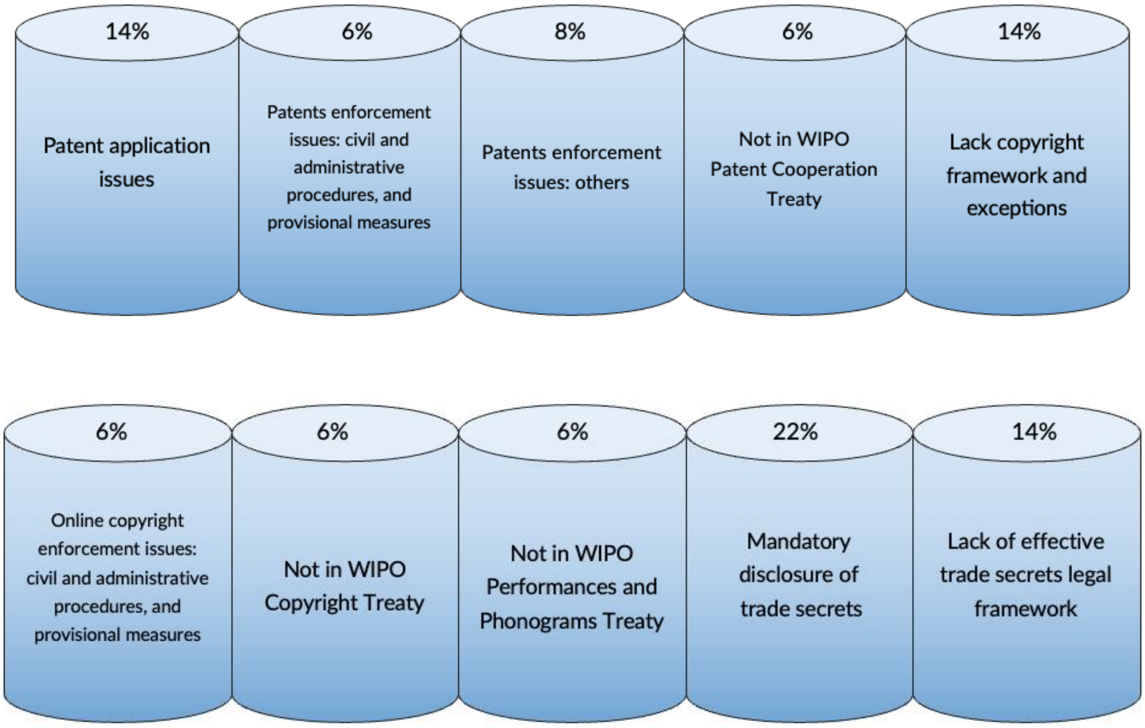
As shown in figure 12, the weight of each intellectual property indicator is given weight of 14%, 6%, 8%, 6%, 14%, 6%, 6%, 6%, 22% and 14%. Mandatory disclosure of business trade secrets without taking a proper measure, the ninth indicator, receives the highest weight of 22% because the disclosure is against the exclusive right. It could risk unfair commercial use and affect the rights holders' commercial value, thereby affecting competition. The indicators with regard to the existence and effective enforcement of patents and trade secrets are assigned an equal weight of 14% because these indicators do not directly take away the rightsholders' competitive advantage. However, without the transparent patent application process, clear copyright exception, and effective patent and trade secrets enforcement could limit innovation and, in some cases, increase infringements.

The remaining indicators captured on the international frameworks and the enforcement of copyright online accounted for the lowest weight of 6% each. The Patent Cooperation Treaty facilitates the patent application process. The Copyright Treaty and the WIPO Performances and

[23] In Singapore, information qualifies as a trade secret if it is not publicly available and is clearly identifiable, as affirmed in Nanofilm Technologies International Pte Ltd v Semivac International Pte Ltd and others [2018] SGHC 167. According to the Intellectual Property Office of Singapore's Trade Secrets Enterprise Guide reports, a breach of confidence is assessed by examining: (1) whether the information is genuinely confidential; (2) whether it was shared under circumstances imposing a duty of confidentiality, even if accessed without consent; and (3) whether the recipient can prove they were unaware of its confidential nature or obtained it accidentally.
[24] See https://www.wto.org/english/res_e/publications_e/ai17_e/trips_art39_oth.pdf

Phonograms Treaty strengthen copyright protection in the digital environment. However, not being a signatory to these treaties does not necessarily violate the protected rights. The inadequate enforcement of online copyright partially focuses on online materials, whereas other indicators capture both online and physical subject matters, therefore having a limited impact.

Figure 12 **Pillar 4 indicators and the weights**



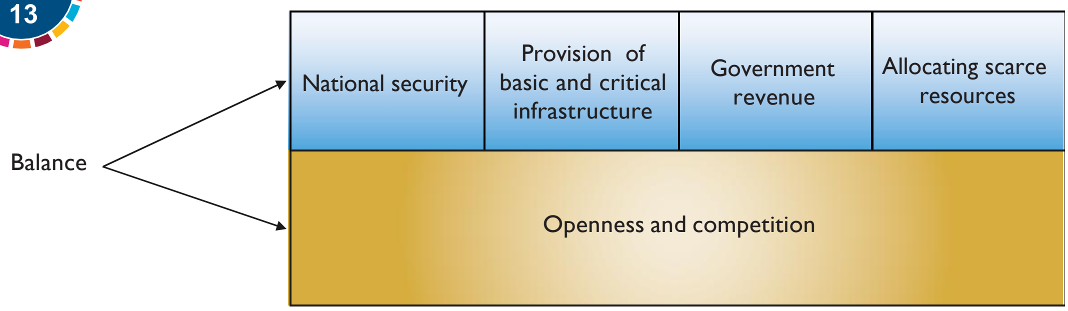
Pillar 5. Telecommunications regulations and competition

Pillar 5 deals with policies regarding telecommunications infrastructure and competition, covering telecommunication services as defined under the WTO GATS telecommunications services. The Pillar includes all forms of basic and value-added telecommunications regardless of the technology used.[25] Basic telecommunications services encompass both public and private services that involve end-to-end transmission of customer-supplier information (such as voice telephone services, circuit-switched data transmission services, and fixed satellite services). Value-added telecommunications refer to services that enhance or add to basic telecom services (such as online data processing, online database storage and retrieval, email, and voice mail).[26] It is important to note that broadcasting services are not covered under this Pillar.[27]

For telecommunications infrastructure and competition policies, relevant policy objectives such as national security, provision of critical infrastructure, raising Government revenue and effective allocation of scarce resources invite regulation of the market by the Government. However, for a domestic telecom market to benefit from digital trade,often by means of foreign investment, these policy objectives need to be balanced, as shown in figure 13, with the policy environment that is conducive to competition among domestic and foreign telecom providers.

Figure 13

Balance in different policy objectives in Pillar 5



Pillar 5 flags telecom policies and practices that undermine the competition in the telecom sector by covering:

- Lack of passive infrastructure sharing;
- Foreign equity limits in the telecom sector;
- Shares owned by the Government;
- Lack of functional/accounting separation;
- Licensing requirements in the telecom sector;
- Not in the WTO Telecom Reference Paper; and
- Lack of an independent telecom authority.

[25] Telecommunication services, available at https://www.wto.org/english/tratop_e/serv_e/telecom_e/telecom_e.htm.
[26] Coverage of basic telecommunications and value-added services, available at https://www.wto.org/english/tratop_e/serv_e/telecom_e/telecom_coverage_e.htm.
[27] The GATS Annex on Telecommunications and the WTO classification (W/120) explicitly differentiate between telecommunication services and broadcasting services. Basic and value-added telecom services involve the transmission of signals but not the content per se, while broadcasting focuses on content delivery.

Useful secondary sources include the International Comparative Legal Guides (ICLG), Lexology ("Getting the Deal Through"), the WTO Trade Policy Reviews Report, the OECD STRI on Telecommunication Services, and the World Bank-WTO Services Trade Restrictions Index (STRI). The secondary sources should only serve to guide researchers to the primary sources.

Lack of passive infrastructure sharing

This indicator looks at (a) whether there is an obligation for passive infrastructure sharing in the economy, and (b) whether the adopted regulatory approach is mandatory or voluntary sharing.

'The passive infrastructure' refers to the sharing of non-electronic infrastructure or the physical infrastructure, such as buildings, sites, network cabinets, masts (towers), poles, ducts and trays.[28] The telecom infrastructure involves high costs, i.e., installation, equipment, operation and maintenance costs. The infrastructure sharing obligation is the process by which one or more telecom operators share infrastructure to deliver services to the end-users. This obligation, thereby, lowers the network deployment, especially in rural areas or marginal markets, stimulates migration to new technologies and the deployment of mobile broadband, and enhances competitionbetween telecom operators and service providers, when safeguards are used to prevent anti-competitive behaviour (ITU, n.d.). Moreover, the infrastructure sharing obligation can be established under various business models and forms.

For example:

- The passive infrastructure sharing obligation is mandated. For example, Thailand requires telecom operators in both fixed and mobile sectors to share their wireless telecommunication network infrastructures, including tower/mast, site, building and other physical facilities, with other telecom operators in a reasonable and fair manner, without discrimination;
- The passive infrastructure sharing obligation is optional, but is implemented. For example, Pakistan, Vanuatu, and Hong Kong (China) do not mandate a passive sharing obligation. However, infrastructure sharing is effectively implemented based on a commercial agreement in both mobile and fixed sectors. New Zealand does not require sharing obligation; however, a framework for voluntary co-location of cellular infrastructure is in place to facilitate sharing of towers, masts and related sites; and
- The passive infrastructure obligation is mandated in specific circumstances or upon request. For example, in Australia, passive infrastructure sharing is practised in mobile and fixed sectors. In the mobile sector, co-location on towers and ducts is negotiated commercially. While in the fixed sector, the network owner of the following declared services, such as line sharing service (LSS), local carriage service (LCS), must provide access to the service upon request.

[28] The infrastructure sharing obligation includes two categories – passive and active infrastructure sharing obligations. The passive infrastructure sharing obligation as mentioned, captures the sharing of physical telecom infrastructure. On the other hand, the active infrastructure sharing obligation captures on active infrastructure of the network (i.e., electronic), including radio access network (antennas, backhaul networks and controllers) and core network (servers and core network functionalities) (GSMA, 2019).

The score is ‘1’ for the absence of a passive infrastructure sharing obligation. The score is ‘0.5’ if the passive sharing obligation is not mandated, but is practiced in the market, or if the obligation is mandated on a case-by-case basis. The score is ‘0’ if the economy mandates at least one obligation for passive infrastructure sharing.

In other words, the steps to take are as follows: **Step 1** check (a) whether there is an obligation for passive infrastructure sharing = if NO =1.0, if YES = can be either 0.00 or 0.5. **Step 2** check (b) whether such a regulatory approach is mandatory or not; if mandatory, score = 0.00, voluntary = 0.5.

Foreign equity limits in telecom sector

This indicator concerns the maximum foreign equity shares in the specific sector of telecommunications. The indicator covers restrictions on foreign equity shares in (a) privately owned companies, and (b) Government-controlled companies—fully or partially SOEs. Foreign equity shares are shares that foreign natural or legal persons hold in a firm incorporated in the investee economy. Foreign investment is a driving force to increase competition and transformation in the telecom market. Limitation on foreign equity shares is a direct obstacle for foreign investors that could induce a higher level of development in the telecom sector. In particular, the foreign equity shares under this Pillar do not include the foreign equity shares in other sectors relevant to digital trade or the e-commerce sector, which are captured in Pillars 3 and 12, respectively.

The score is ‘1’ if there is a ban on foreign ownership in the telecommunication sector or if only a minority stake (less than 50%) is allowed in more than one measure. The score is ‘0.8’ if only a minority stake is allowed. The score is ‘0.5’ where a controlling stake (more than 50%) is allowed. The score of ‘0.5’ is also applied where limitations on foreign equity shares only existin SOEs within the telecommunication sector. The score is ‘0’ if there is no limitation on foreign equity share in the telecommunication sector.

Shares owned by the Government

This indicator concerns the percentage of shares owned by the Government in telecom companies, including the case in which the telecom operator is an SOE. Most of the Government ownership in the telecommunication company limits the local and foreign ownership in the telecom market.

When an SOE is operating in the telecommunications sector, it is important to identify (a) the percentage of Government ownership, and (b) the company’s market share expressed as a percentage to assess the presence of the SOE and potential impact on the respective economy’s telecom market. Telecommunications market structures vary, for example:

- A de jure monopoly (statutory monopoly) is in place when telecommunications networks and services must be exclusively state-owned. For example, Turkmenistan’s Law on Communication mandates exclusive state ownership of telecom networks used for national security and national broadcasting. Turkmentelekom TC and Altyn Asyr CJSC (TMcell) are 100% state-owned;

- A de facto monopoly exists when an SOE or private company is the sole operator in a market but absence of a legal barrier preventing competition;
- Government ownership through a foreign SOE is when the national telecom operator is owned by a foreign SOE, not a local Government. For example, Amalgamated Telecom Holdings Kiribati Limited (ATHKL) is the sole telecom provider in Kiribati. The company is a subsidiary of Fiji-based Amalgamated Telecom Holdings (ATH), which majority majority-owned by the Fijian Government;
- Partial Government ownership without market dominance refers to when the Government owns shares in a telecom company, but private operators dominate the market. For example, in Thailand, market dominance differs by segment. Although the fixed line telephone market is dominated by the National Telecom, SOE, private operators, Advanced Wireless Network (AWN), and True Move H Universal Communication (TUC) are dominant providers in the mobile market; and
- Multiple SOEs operating in a competitive market is when more than one SOEs operate and compete with other SOEs and/or private operators. For example, China United Network Communications Group (China Unicom), China Telecom Corporation Limited (China Telecom), and China Mobile Limited are wholly owned SOEs providing basic, mobile and value-added telecommunication services.

The score is ‘1’ if the Government’s ownership in at least one company is above 50%. The score is also ‘1’ if Government ownership in more than one company is between 1% and 50%. The score is ‘0.5’ if Government ownership in one company is between 1% and 50%. The score is‘0’ if there are no shares owned by the Government in telecom companies.

This information, particularly on market share and ownership structures, can be found in telecom market reports or annual updates published by the telecom regulator or Government authorities responsible for the telecommunications sector. Official websites of telecom companies can also serve as a useful source.

Lack of functional/accounting separation

This indicator asks whether the economy mandates functional and/or accounting separation for operators with significant market power (SMP) in the telecom market. The SMP refers to the regulatory status representing a dominant position in the telecom market. Functional and accounting separation enhances cost transparency, promotes fair market prices, and avoids SMP and non-discriminatory practices in telecom markets.

‘Functional separation’ (sometimes known as ‘operational separation’) refers to the separation of units operating different activity branchesin the telecommunication company. The functional separation prevents entities with a domain position from controlling operations in another area. For example:

- In Georgia, the National Communications Commission is empowered to identify operators with SMP through market analysis and may require them to separate functional resources within their networks; and
- New Zealand implemented structural separation requiring Telecom New Zealand to separate its network operations from its retail services. Accordingly, the company was split into two entities, which are Chorus, which operates the physical telecom infrastructure, and Spark (formerly Telecom), which provides retail services.

'Accounting separation' refers to the separation of accounting records for different businesses and parts of businesses run by the same company, so that the costs, revenue and assets associated with each part of a business can be separately identified and properly allocated. This kind of separation ensures that the telecom company accurately reports its financial performance in each area of its operations. For example:

- In Armenia, operators with SMP shall maintain separate accounts for different services. Similarly, Australia mandated a telecom company to maintain separate accounts for its different technology units; and
- Pakistan requires a telecom service provider in both fixed and mobile services that is determined as having an SMP in the market to submit the costs and revenues for each regulated service, including profit and loss statement, balance sheet information, return on capital employed and supporting notes.

The score is '1' for the absence of separation requirements. The score is '0.5' if only accounting separation. The score is '0.25' when only functional separation is mandated. The score is '0' if the economy mandates both accounting and functional separation of dominant network operators.

Licensing requirements in telecom sector

This indicator refers to a license for private telecommunication services or a license to operate telecommunication facilities. It asks whether the licensing requirements include (a) discriminatory conditions against foreign firms, or (b) create significant barriers to entry for both foreign and local service providers. These may include mandatory performance requirements, separate licensing for value-added telecommunication services, caps or limits on the number of licenses issued to obtain licenses for providing telecom facilities or services, and capital restrictions. For example:

- Indonesia carries out administrative licensing to allocate radio frequencies rather than holding an auction for the frequencies;
- Bangladesh mandates operators of nationwide telecommunication transmission networks to share a portion of their revenue with the regulator starting from the second year of operation, in addition to license acquisition and annual fees;
- Tanzania requires holders of licenses for network facilities and network services to offer a minimum of 25% of the company's share to the public through an initial public offering on the stock market; and
- Other requirements, such as separate licensing requirements for value-added telecommunication services, and caps or limits on the number of licenses issued.

A license scheme with conditions that include discriminatory conditions against foreign firms or create significant barriers to entry for both foreign and local service providers counts as '1'. Otherwise, the score is '0'.

Useful secondary sources include the WTO Trade Policy Reviews Report, the Freedom House's report on Freedom on the Net, the WorldMap of Encryption and the Lexology ("Getting the Deal Through"). The secondary sources shouldonly serve to guide researchers to the primary source

Not in the WTO Telecom Reference Paper

This indicator captures whether an economy is a signatory to the WTO's Telecom Reference Paper. The Reference Paper prescribes definitions and principles of the six regulatory frameworks for the basic telecommunication services on competitive safeguards, interconnection, universal services obligation, public availability of licensing criteria, independent regulators, and allocation and use of scarce resources. The regulatory framework is legally binding for those WTO Governments which have committed to it by appending the document, in whole or in part, totheir schedules of commitments and is enforceable through the WTO dispute settlement (box 5).

The score is '1' if an economy is not appended to the WTO Telecom Reference Paper. The score is '0' if an economy is fully or partially appended to the WTO Telecom Reference Paper. To see the coverage schedule, check the list of telecommunications commitments and exemptions at the WTO schedule of specific commitments and lists of Article II exemptions, telecommunications commitments and exemptions[29] or the official government website.

Box 5

WTO Telecom Reference Paper

In general, all WTO members are bound to GATS which incorporates an Annex on telecommunications to ensure reasonable access and the use of public telecommunications. The WTO Telecom Reference paper is another key element of telecom disciplines resulting from the post-Uruguay Round Ministerial Decision on negotiations of basic telecommunications. The Reference Paper has the status of international treaty. It introduces the regulatory component that allows WTO members to commit to this framework either wholly or partially by appending the document to their schedules of commitments. The purpose of this regulatory framework is to provide a blueprint considering a set of best practices for telecommunications reform when competition is being introduced in the market.

[29] When an economy does not appear in the 'WTO telecommunications commitments and exemptions' list, it does not always mean that an economy does not append to the WTO Telecom Reference Paper. Please double check at the WTO schedule of specific commitments and lists of Article II exemptions or the official government website.

Lack of independent telecom authority

This indicator asks whether an economy has an independent telecommunications regulator.[30] It looks at the existence of an executive authority for the supervision and administrationof services in the telecommunications sector that is completely independent from the operator and supplier of the telecommunication services, the Government and other interested persons in the decision-making process and administration of decisions. An independent telecom authorityis expected to promote fair competition and not be involved in a conflict of interest. This includes two dimensions of independence, which are functional independence and institutional and financial independence.

'Functional independence' refers to the regulator's ability to carry out its duties and make decisions impartially without interference from interested parties:

- The regulator can make decisions without approval from Government bodies or telecom suppliers. While decisions may be overruled by higher authorities, such mechanisms should be clearly defined, limited in scope and not used to undermine impartial regulation.
- Regulatory procedures are applied equally to all market participants. There is no preferential treatment given to specific companies, such as SOE or dominant firms.
- While it may be established under or report to a ministry, the regulator functions independently and is not subject to political influence in its operation.

'Institutional and financial independence' refers to the regulator's organization and financial autonomy from telecom operators and Government:

- The regulator is legally separate from the telecom service provider. It does not own, manage or control any telecom company.
- The regulator has a dedicated budget, funded in a stable and predictable manner (e.g., through industry levies or a stable parliamentary appropriation), and is not subject to arbitrary or undue budget control by the Government that could compromise its operational autonomy.
- The regulator is authorized to recruit, appoint or manage its staff without undue interference from Government (beyond established high-level appointments) or telecom operators.

The score is '1' for the lack of an independent telecom authority. Otherwise, the score is '0'.

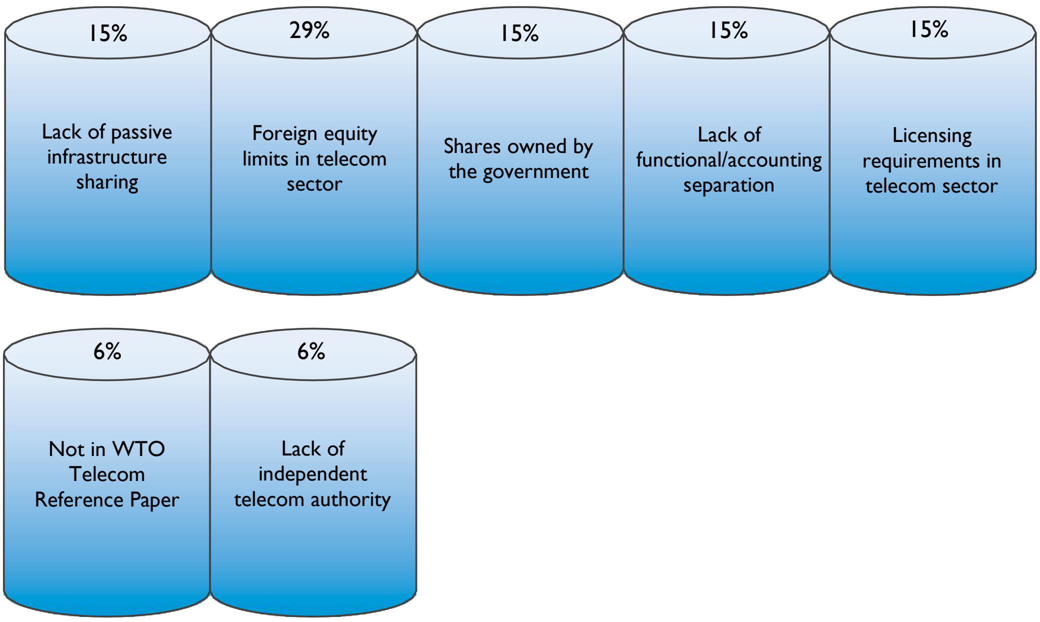
The weights for each indicator

As shown in figure 14, each indicator is given the weight of 15%, 29%, 15%, 15%, 15%, 6% and 6%. The indicators on foreign equity shares and shares owned by Governments reflect the structure of the telecom market in the respective economy and capture the presence of competition in the telecom market. Other indicators on passive infrastructure sharing obligations, functional/accounting separations, licensing schemes that the WTO Telecom Reference Paper and independent telecom authority provide the extent to which competition is in place in a certain telecom market.

[30] The concept of independent telecom authority is embodied in the WTO Telecom Reference Paper (Clause 5 Independent Regulators), and the WTO Reference Paper on Services Domestic Regulation Section 2, Clause 12 Independence). Specifically, the WTO Reference Paper on Services Domestic Regulation clarifies that "For greater certainty, this provision does not mandate a particular administrative structure; it refers to the decision-making process and administering of decisions. "In the RDTII 2.1, the scope of independent telecom authority is beyond the WTO Reference Paper for the independence from telecommunication operators; it covers the independence from the Government and other interested parties.

Foreign equity limits, the second indicator, is assigned the highest weight of 29%. Foreign ownership limitations discriminate against foreign investors and preclude foreign investment in the telecom sector. For the third indicator, the higher percentage of shares owned by Governments directly affects the competition in the telecom market. However, it affects local and foreign businesses, thereby receiving a lesser weight of 15%. The lack of passive infrastructure sharing requirements and the lack of functional and/or accounting separations, the first and fourth indicators,are assigned a similar weight of 15% because the absence of these measures could hamperthe entry of new players into the market and discourage the telecom operators from competing with the established operators. In other words, the lack of measures could provide the operators with the dominant position's ability to engage in anti-competitive practices by increasing their control over the telecom-related facilities. The strict licensing requirements, the fifth indicator, potentiallyblock or discourage businesses from participating in the telecom sector, hence receive the scoreof 15%. Moreover, for the sixth and seventh indicators, the WTO Telecom Reference Paper and independent telecom authority are given the least weight of 6% since the absence of these indicators does not mean that competition does not exist, and the extent to which these measuresdeter foreign business is somewhat minimal.

Figure 14 **Pillar 5 indicators and the weights**



Pillar 6. Cross-border data policies

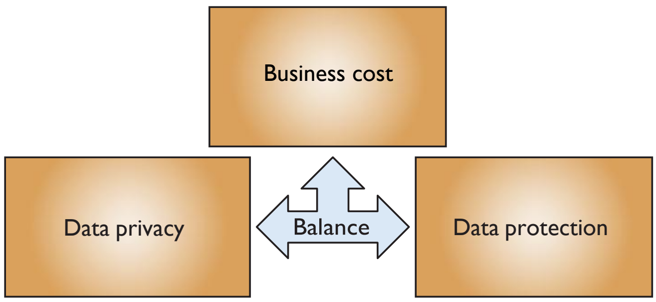
Pillar 6 deals with cross-border data policies by regulating the ways in which data flows from one jurisdiction to another. Important policy considerations are about balancing business costs created by the regulations with the public policy objectives to protect data and data privacy(figure 15).

Regarding business costs, the regulation of cross-border data flows tends to increase the cost of compliance as it sets barriers for businesses to store and process data (Ferracane, 2017). Transferring data across borders is a crucial driver of digital trade, as data is integral to the provision of digital goods, online services and even digital-trade infrastructure. Specifically, business models in these areas rely on 'data value chains'. A data value chain, by connecting data acquisition, data storage, data processing and data analysis, offers efficient and smart business solutions for transactions. These transactions occur either within a business or betweena business and its customers. Therefore, barriers to the movement of data across borders could heighten the costs for digital trade.

Data privacy and data protection are two sides of the same coin. Data privacy refers to an individual's right to retain control over the way in which their personal data is collected and used, while data protection refers to the responsibility of entities to apply safeguard mechanisms to the handling of data (PECC and Access Partnership, 2021). Without proper data protection, data privacy is threatened; therefore, a high level of data privacy often presupposes strong data protection.

Sound cross-border data policies are often based on a subtle balancing of business costs and digital trust (figure 15). Compliance with data protection rules increases costs. However, the better that data privacy is, and the stronger that cybersecurity is, the greater digital trust the regulatory environment evinces in the eyes of businesses and consumers. This is because there would be fewer data breaches, and, even if there were, stronger accountability mechanisms would exist.

Figure 15 **Balance in different policy objectives in Pillar 6**



Pillar 6 covers regulatory measures (or lack thereof) on cross-border data transfer that do not get justified in the light of these policy parameters, not necessarily because the measures omit any one of the policy parameters, such as business cost, data privacy and cybersecurity, but because the measures fail to find a proper balance of these objectives. The failure to lie in such a balance point makes the measures possibly discourage businesses from engaging in digital trade in the respective regulatory environments.

These regulatory measures (or lack thereof) that Pillar 6 flags tend to be more costly if they apply to personal data rather than non-personal data. **Personal data** refers to any information that relates to an identified or directly or indirectly identifiable individual. Per this definition, personal data are generally (a) sensitive data such as name, surname, email address, identification card, IP address, cookie ID as well as health-related data and data revealing racial or ethnic origin, beliefs and religion, and (b) pseudonymous or 'de-identified' data, i.e., data that make an individual identifiable with additional information. By contrast, **non-personal data** are anonymous data that do not relate to an identified or identifiable individual.

Overregulating personal data flows has a higher opportunity cost than overregulating non-personal data flows. Personal data is the basis of cross-border online services such as financial, business and IT services. Companies analyse the personal data of their clients to offer the services. Furthermore, personal data creates an opportunity for enterprises to improve their consumer engagement for online services or other types of digital trade (Anant and others, 2020). Personal data, such as location data, websites browsed, searches performed, apps and programs used and Internet usage times, allows companies to understand consumers' needs better. These insights, in turn, help to develop new products and services as well as to personalize advertising and marketing. In addition, personal data can often be anonymized or aggregated to become non-personal data, allowing its use for analytical and innovative purposes without privacy concerns. Over-regulating personal data prevents this "unlocking" of value.

This Pillar evaluates the regulatory environment for cross-border data flows through the following indicators:

- Ban and local processing requirements;
- Local storage requirements;
- Infrastructure requirements;
- Conditional flow regimes; and
- Not in agreement with binding commitments on data transfer.

In this Pillar, the requirements on location of data and data flows generally are found in comprehensive data laws or sectoral laws governing the health sector, financial sector (e.g., credit card information) and telecommunications sector (e.g., computer traffic data), for example. Useful secondary sources include the OECD STRI database and specialized databases, such as Linklaters, DataGuidance, Lexology and DLA Piper. The secondary sources should only serve to guide researchers to the primary sources.

Ban and local processing requirements

This indicator asks whether there is a ban on data transfer and a local processing requirement. As the business costs arising from a ban and local processing requirements are quite subtle, these requirements are classified under the same indicator (Ferracane, 2017).

'Ban on data transfers' prohibits, *per se*, cross-border transfer of data. **'Local processing requirement'** mandates that businesses process certain data domestically. Processing data refers to various activities involving data, such as collection, organization, structuring, storage, adaptation, use, disclosure and dissemination (European Commission, 2018a). Moreover, to process data locally, foreign companies often need to hire domestic service providers even though the companies may already have their own data processors. This constitutes additional costs for them. The following are examples of local processing requirements:

- Australia requires health records that contain personal or identifying information to be stored and processed locally;
- The Republic of Korea requires financial service providers that use cloud services to locally process credit and unique identification information of their users; and
- Thailand requires a credit information company, information controller, or information processor operating business in Thailand to control or process information locally.

The scoring metrics of this indicator have three features: (a) type of data, a requirement that applies to personal data will get a higher score than a condition that applies to non-personal data; (b) scope of data, a horizontal requirement that applies across sectors will get a higher score than a requirement that applies only to a specific sector (such as financial services or telecommunication sector) or specific data types (such as accounting data and health records); and (c) number of economies, a requirement that applies to a greater number of economies will get a higher scope than a requirement that applies to one economy. A measure applied to the Government data should not be scored. The indicator focuses on a measure potentially affecting commercial transactions.

The score is '1' when a ban and/or a local processing requirement covers personal data or applies horizontally across sectors. The score of '1' is also assigned when there are two or more requirements on ban and/or local processing applied to non-personal data, or a specific set of data, or applies to more than one economy. The score is '0.5' when a ban and/or a local processing requirement applies to non-personal data, or a specific set of data, or to one economy.The score is '0' when the data is permitted to be transferred freely without any requirement.

Local storage requirements

This indicator asks whether there is a local storage requirement. **'Local storage requirement'** mandates that a copy of certain data be stored within the economy. Businesses can transfer data across borders as long as a copy of the data is kept within the economy. The following are examples of local storage requirements:

- Australia requires health records that contain personal or identifying information to be stored and processed locally;
- Malawi implements a local storage requirement for health-related data;
- New Zealand requires registered entities to store specified tax-related records locally;

Notably, many countries implement both requirements for sensitive data, like personal and financial information. However, some regulations may only require that data be processed locally to ensure it is subject to a nation's laws, while still allowing the data to be stored elsewhere after processing is complete. Hence, in the case that the law requires both local processing and local storage, the measures will be recorded in both places, as in the example of Australia.

- Salvador requires legal entities operating as data information agencies to maintain their databases in the country and to allow access to the superintendence of the financial system; and
- Türkiye requires both domestic and foreign social network providers with over one million daily users in Türkiye to store user data within the economy.

Some examples (such as Australia) show how some economies impose both local storage and local processing requirements under the same measure. However, the local processing requirements are more demanding than the local storage requirements because a mandate to process data locally typically necessitates local storage, while a local storage requirement does not necessarily mandate local processing, thus potentially raising the cost more significantly for firms.

The scoring metrics of this indicator have two features: (a) type of data – a requirement that applies to personal data will get a higher score than a condition that applies to non-personal data; and (b) scope of data – a horizontal requirement that applies across sectors will get a higher score than a requirement that applies only to a specific sector (such as financial services or telecommunication sector) or specific data types (such as accounting data and health records). A measure applied to Government data should not get scored. The indicator focuses on a measure potentially affecting commercial transactions.

The score is ‘1’ when a local storage requirement covers personal data or applies horizontally across sectors. The score is ‘1’ and is also assigned when there are two or more local storage requirements applied to non-personal data or a specific set of data. The score is ‘0.5’ when a local storage requirement applies to non-personal data or a specific set of data. The score is ‘0’ when the data are permitted to be transferred freely without any requirement.

Infrastructure requirements

This indicator asks whether there is an infrastructure requirement. **‘Infrastructure requirement’** mandates the establishment of a local data centre as a condition to provide certain services using data.

It is important to note that an infrastructure requirement is different from a local storage requirement (indicator 6.1). Specifically, the local storage requirement only mandates that data be stored within the economy but does not require the service provider to build or own a data centre; the provider may rent or use an existing local facility or server. By contrast, infrastructure requirement requires the service provider to establish or use dedicated local physical infrastructure as a precondition for offering services. The following are examples of infrastructure requirements:

- Chile requires banking institutions to outsource data processing services outside the country to have a contingency data processing centre located within the country.
- Mongolia requires servers processing sensitive, biometric and genetic data to be physically located within the economy and comply with technical and security standards;
- Kazakhstan requires operators of communications networks to establish a local system of centralized management for their networks; and

- Viet Nam requires providers of websites, social networks, mobile networks and online games to maintain at least one local server.

The score is ‘1’ when there is at least one infrastructure requirement. Otherwise, the score is ‘0’. A measure applied to the Government data should not be scored, because the indicator focuses on a measure potentially affectingcommercial transactions. Additionally, the indicator focuses on the free flow of data, specifically looking for requirements that act as a primary barrier to data movement, such as regulations that explicitly mandate the establishment of a data center or other local data storage infrastructure as a condition for service. The indicator does not score operational or technical requirements that apply to data centers.

Conditional flow regimes

This indicator asks whether an economy adds other conditions for cross-border data transfer. These are another set of conditions that businesses and organizations need to satisfy to transfer data across borders. In other words, even if businesses and organizations satisfy the local storage or processing requirements, they do not get to transfer data unless they also satisfy the set of conditions that this indicator deals with (assuming that the economy imposes these conditions). These conditions prevent businesses and organizations from transferring data to economies where the level of data protection is not adequate or equivalent to the level of domestic data protection.

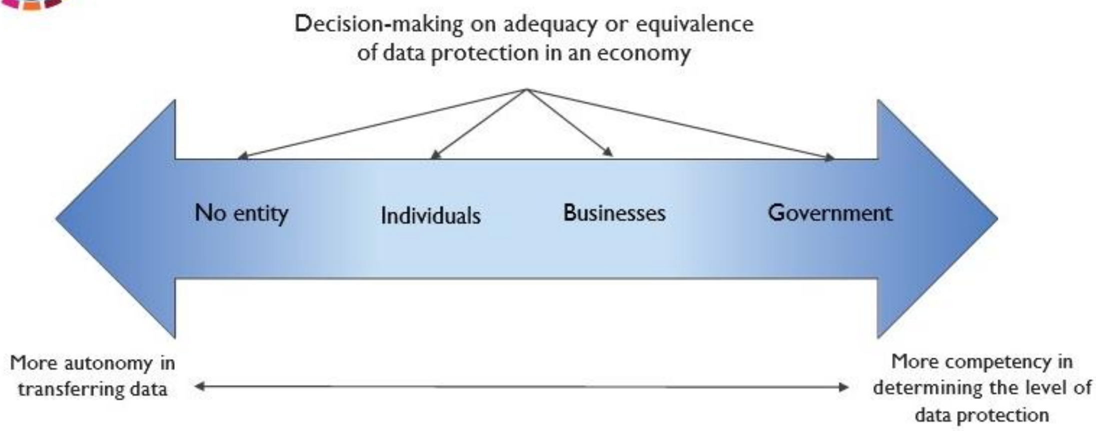
The conditions vary depending on which entity decides whether a particular economy has that requisite level of data protection (OECD, 2018). For example:

- There is no condition attached to cross-border data transfer. Thus, naturally, no entity decides the question of the adequacy and equivalence of data protection in economies where data gets transferred. In this case, economies often allow data flows freely across borders, assuming that businesses and organizations that transfer the data are held accountable and liable for possible data breaches that take place in the destination economies;
- Data subjects, who are often individual users, decide whether to allow the transfer of their data to a particular economy by providing their consent;
- Businesses and organizations decide it by evaluating whether a particular economy to which data are about to be transferred has an adequate or equivalent level of data protection;
- The Government determines that certain economies have that requisite level of data protection.

However, to determine which one of these conditions is more costly than the other is difficult. On the one hand, the lack of such a condition, the consent mechanism, and the mechanism for businesses and organizations' evaluation on this spectrum of conditions, as shown in figure 16, seems to restrict the movement of data less than the mechanism for the Government's approval. This is either because the movement of data lies in the autonomy of individuals (i.e., data subjects) or because the decision-making is reserved for businesses and organizations rather than the Government. On the other hand, individual users or even businesses may not be competent entities for determining which economy has an adequate or equivalent level of data protection; rather, it is the Government that is in a better position to determine that question, i.e., which economy has an adequate level of data protection.

Figure 16

Conditions of consent, evaluation and approval



The following are examples of these conditions:

- Brazil sets out regulations on how financial institutions and other institutions regulated by the Brazilian Central Bank should hire cloud computing services from providers that store or process information outside Brazil. In the absence of a formal agreement with the regulators of the economy where the services are performed, prior authorization is required at least 60 days in advance;
- Palau requires agencies to notify the Ministry of State and obtain individual consent before transferring personal information abroad. The Ministry of State shall determine whether to prohibit a transfer based on its impact on individuals, the importance of facilitating free information flow and relevant international guidelines or treaties;
- The Republic of Korea requires permission from the Minister of Land, Infrastructure and Transport for the transfer abroad of geographical data related to maps or photos produced for survey purposes;
- Singapore requires that the transfer of personal data abroad comply with the Personal Data Protection Act (PDPA) obligations. Recipients outside the country must obtain individual consent to transfer the data and provide a comparable standard of personal data protection as provided in the PDPA; and
- Uzbekistan allows cross-border personal data transfers if the foreign state provides adequate protection, the data subject consents, the transfer protects public order or citizens' rights, or if required by international treaties. The transfers of personal data may be restricted in order to protect the constitutional system, the rights and legitimate interests of citizens or to ensure national security.

Accordingly, this indicator differentiates the conditions, not based on types of decision-making entities, but based on the two features: **(a) type of data**, a requirement that applies to **personal data, will get a higher score than a condition that applies to non-personal data**; and (b) **scopeof data, a horizontal requirement that applies across sectors will get a higher score than a requirement that applies only to a specific sector** (such as financial services or telecommunication sector) or specific data types (such as accounting data and health records).

Therefore, the score is '1' if the regime covers **personal data** (regardless of whether it is horizontal or sector-specific). The score is also '1" if the conditional flow regime applies horizontally across all sectors (even if it only applies to non-personal or specific data types). The score is '0.5' when a conditional flow regime applies to non-personal data and/or a specific setof data (e.g., financial, telecommunications, cloud services, etc.). The score is '0' when there isno condition. Moreover, Government data are not listed because the database captures commercial activities.

Not in an agreement with binding commitments on data transfer

This indicator asks whether an economy has committed itself to agreements with a binding commitment on transferring data across borders. The agreement with a binding commitment captures all forms (preferential trade agreements or other interoperability initiatives, such as treaties and regional agreements) and types (bilateral, plurilateral, or multinational). The binding provision is enforceable and obliges the party to comply with the agreed provisions. Participating in binding agreements or initiatives safeguards the flow of data and lowers compliance costs.

As for trade agreements, the cross-border data flows provision generally focuses on the commitments to transfer information, including personal information, by electronic means. The data flows provision can be found under various denominations, for example, 'Cross-Border Transfer of Information by Electronic Means', 'Cross-Border Information Flows', 'Movement of Information' and 'Free Flow of Data'.

For scoring, the score is '1' if an economy does not commit to any agreements with a binding commitment [31] on cross-border data transfer. The score is '0' if an economy signs at least one binding agreement on data flows. In case the data flows provision or treaty contains both binding and non-binding obligations (considered mixed legalization), the score will be 1. It is important to review exceptions clauses and special and differential treatment granted to the economy within the agreement to determine the scope of the economy's commitment.

For this indicator, trade agreements, treaties and other commitments are the primary sources, such as the official text of the agreements. The Trade Agreements Provisions on Electronic-commerceand Data (TAPED) dataset 2.2.1 [data_free-flow_prov] is a useful secondary source. The secondarysources should only serve to guide researchers to the primary sources.

The weights for each indicator

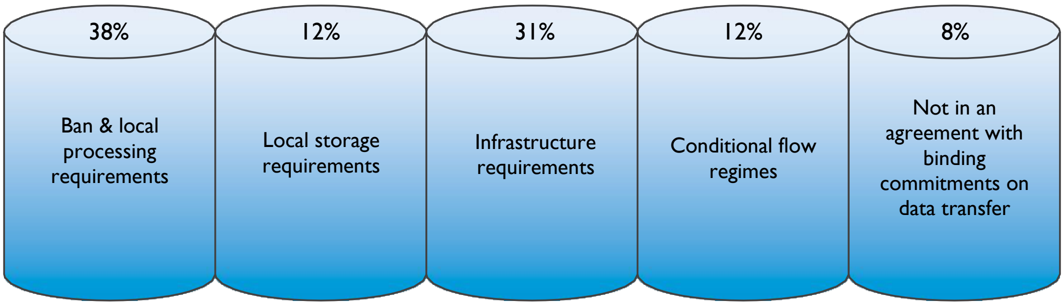
As shown in figure 17, the weight of each indicator varies with regard to cross-border data flows at 38%, 12%, 31%, 12% and 8%, respectively. The first and the third indicators – ban and local processing requirement, as well as infrastructure requirement – are regarded as more weight than the other indicators, at 38% and 31%. A ban on data transfer is costly, in that data is a crucial driver for digital trade. This requirement prohibits data flows, while the rest of the data requirements and conditions permit the data to transfer freely once they have been fulfilled. Local processing requirements relate to several data activities, and businesses may feel a need to build local servers or hire domestic service providers, even though these are not mandatory. Under the infrastructure requirements, the businesses are bound to establish data centres or local servers within the economy, increasing fixed costs. Although the requirements to process or build an infrastructure for data tend to incur a higher cost of compliance, the processing requirement could create a higher cost as it involves more activities.

Local storage requirements and conditions for cross-border data receive an equal weight of 12%. Local storage has a lower compliance cost because merely a copy of the data is required to locate within the territory. Still, the requirement may not necessarily promote digital trust becausethe local storage does not address how data subjects' data is actually used abroad. The condition on data flows, while it incurs costs against businesses, has a rational relationshipwith ensuring digital trust in regulatory environments where data is to be used, although tangential. For example, the condition that data cannot be transferred unless data protection in receiving economies is deemed adequate tends to ensure that data gets some protection there. However, since the regulating economy has no or little control over data privacy and protection, the effect of such a condition on forming a digital trust may be limited.

[31] The extent of legalization, i.e., soft legalization (non-binding obligation), mixed legalization (binding and non-binding obligation) and hard legalization (binding obligation), see the latest version of the TAPED Codebook, available at https://www.unilu.ch/en/faculties/faculty-of-law/professorships/burri-mira/research/taped/. This indicator is based on TAPED data_free_flow_prov_2_2_1, which looks at a dedicated provision on data flows.

In addition, the last indicator, participation in agreement with binding commitment on cross-border data flows, has the least weight (8%). Committing to a binding agreement could help in shaping harmonize data flows regulation and practice across regions. However, not being a member of data flows commitment does not impede the flows of data across borders.

Figure 17 **Pillar 6 indicators and the weights**

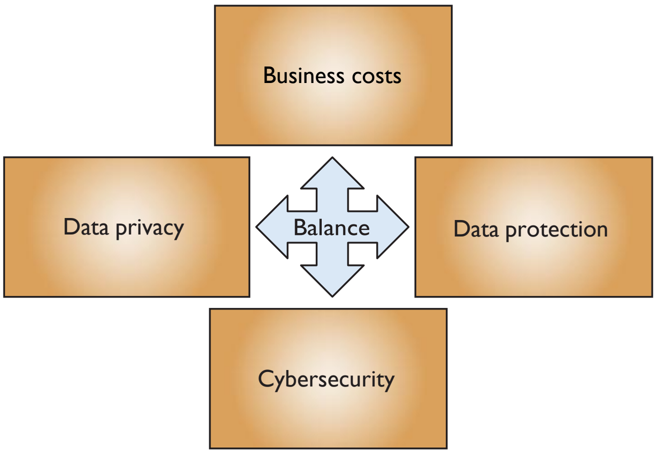


Pillar 7. Domestic data protection and privacy

Pillar 7 deals with policies governing the use of data in the regulating economy. In addition to the three policy parameters – business cost, data privacy and data protection – this pillar also focuses on cybersecurity (figure 18). Cybersecurity refers to a Government's enforcement efforts to protect organizations, individuals and networks, both in the public and private sectors, from digital attacks such as "the unauthorized access, modification and extraction of data, the theft of proprietary information and the purposeful incapacitation of critical infrastructure, depending on the scale and intention of the attack in question" (PECC and Access Partnership, 2021). Along with data privacy and data protection, cybersecurity has a significant impact on digital trust.

Figure 18

Balance in different policy objectives in Pillar 7



This Pillar considers the following conditions as potentially creating high costs:

- Lack of a comprehensive legal framework for data protection;
- Lack of a dedicated legal framework for cybersecurity;
- Minimum period of data retention requirements;
- Data Protection Impact Assessment (DPIA) or Data Protection Officer (DPO) requirements;
- Requirements to allow Government access to personal data.

Useful secondary sources include the UNCTAD Cyberlaw Tracker, Linklaters, DataGuidance, ICLG, Lexology and DLA Piper. The secondary sources should only serve to guide researchers to the primary sources.

Lack of comprehensive legal framework for data protection

This indicator asks whether an economy has adopted a comprehensive data protection legal framework that applies to personal data across sectors. The significance of data privacy protection lies in its contribution to digital trust and assurance of legal certainty. In this regard, the lack of data protection laws will create risks for the data subject and pose challenges for businesses. Specifically, the absence of a data legal framework could discourage digital trade becausethe data users become hesitant to entrust their data with businesses on how their rights will be enforced, and the willingness to transfer them to other economies. Businesses and organizations can be held liable for the consequences of data breaches.

The criteria for a **'comprehensive'** data protection legal framework include broad, cross-sectoral (i.e., horizontal) applicability and detailed provisions on the scope and application of rights and obligations of data subjects. The comprehensive data protection legal framework empowers individuals to controltheir personal data, such as the right to access, rectification, erasure, and data portability, and encompasses various activities, such as data collection, data processing and the transferof personal data across borders.

Compounding this problem is the fragmentation of data protection requirements across sectors. Sectoral data protection obligations, such as in finance, health and education, can be interoperated (PECC and Access Partnership, 2021). Often, several sectors are entwined with each other. As seen in the rise of FinTech, data privacy threats to communications and IT sectors also constitute threats to the finance sector. The education, health and e-commerce sectors are dependent on the payments sector, which is, in some jurisdictions, also categorized as a financial entity. Thus, the asymmetry of information from different data privacy and data protection requirements across sectors may add compliance costs to businesses as they navigate through the complexities of regulations.

In this context, regional agreements encourage establishing personal data protection legal frameworks and ensure data privacy protection while facilitating the free flow of data. This facilitation often includes mutual recognition of agreement partners' data protection certifications or data protection trustmarks[32] as a mechanism to transfer data.[33]

Accordingly, the scoring metrics for this indicator consider that if an economy lacks this data protection framework, the score is '1'. The score is '0.5' when there is a data protection framework that applies only to specific sectors ('sectoral law'). The score is '0' if there is a comprehensive data protection legal framework.

[32] Data protection trustmarks are a voluntary certification for organizations to demonstrate accountable data protection practices following the international frameworks, such as the European Union's General Data Protection Regulation (GDPR) and the APEC Privacy Framework. These trustmarks promote competitive advantage from business and build trust for consumers (Singapore, 2020; Singapore, 2024).

[33] Specifically, CPTPP Article 14.8, and DEPA Article 4.2 state that "to this end, each Party shall adopt or maintain a legal framework that provides protection of the personal information of the users of electronic commerce and digital trade…Each Party shall endeavour to adopt non-discriminatory practices in protecting users of electronic commerce from personal information protection violations occurring within its jurisdiction." Notably, DEPA further emphasizes the interconnectedness between data protection and the free flow of data: "The Parties shall endeavour to mutually recognize the other Parties' data protection trustmarks as a valid mechanismto facilitate cross-border information transfers while protecting personal information." Similarly, the African Union Convention on Cyber Security and Personal Data Protection Article 8.1 states that "Each State Party shall commit itself to establishing a legal framework aimed at strengthening fundamental rights and public freedoms, particularly the protection of physical data, and punish any violation of privacy without prejudice to the principle of free flow of personal data."

Lack of dedicated legal framework for cybersecurity

This indicator asks whether an economy has adopted a dedicated cybersecurity legal framework. A cybersecurity legal framework ensures computer security incident response and potentially discourages cybercrime, such as malicious intrusions or the dissemination of malicious codes that affect the electronic networks. Therefore, the presence of a cybersecurity legal framework ensures digital trust and legal certainty.

The criteria for a **'dedicated'** cybersecurity legal framework refer to when an economy establishes broadly applicable cybersecurity-specific laws (horizontal) or cybersecurity-focused laws applicable to specific sectors (sectoral). A dedicated cybersecurity legal framework strengthens enforcement against cybercrime offences. Whereas a "non-dedicated" cybersecurity legal framework refers to an economy that does not implement specific cybersecurity laws but relies on other laws to govern threats arising from cybercrime. This includes laws applicable to the monitoring, detection, prevention, mitigation, and management of incidents.

For scoring, if an economy lacks a cybersecurity legal framework, the score is '1'. The score is '0.5' when there is a non-dedicated cybersecurity legal framework and/or a dedicated cybersecurity legal framework that applies to specific sectors. The score is '0' if a dedicated cybersecurity legal framework applies to all sectors. Useful secondary sources include ICLG Cybersecurity Laws and Regulations, Baker McKenzie Global Data Privacy and Cybersecurity Handbook, and UNCTAD Cybercrime Legislation Worldwide. The secondary sources should serve only to guide researchers' attention to the primary sources.

Minimum period of data retention requirements

This indicator asks whether an economy imposes a minimum period of data retention requirements. The data retention requirements regulate how long a company should keep a copy of certain data in order and make it available upon request by the authorities.[34]

The purpose behind these requirements is often to help investigations on certain matters, such as corporate affairs and tax payments or to reinforce the Government's law enforcement efforts, especially with regard to communication data from telecom companies.

However, these requirements can pose challenges to achieving a balance among important policy objectives for digital trade – namely, business cost, data privacy and cybersecurity. The data retention requirements increase compliance burdens on businesses. In particular, MSMEs often lack resources to manage data they keep for a substantial period of time per divergent regulatory obligations across economies. The availability of data retained for the purposes of various regulatory investigations might create the false impression that data retention is necessary. The mere convenience of data retention does not necessarily make it necessary.

The requirements also undercut data privacy by requiring companies to store personal data for a set period, even longer than necessary. In some cases, the data retention schemes have become a Government tool for accessing personal data (Rucz and Kloosterboer, 2020). Last, the data retention requirements ironically and potentially weaken cybersecurity and digital trust, because data retention practices increase data security risks, including data leaks, abuses and misuses. For example, potential unauthorized disclosure of, or access to, retained telecommunications data

[34] Data retention requirements differ from local storage requirements in Pillar 6. The retention requirements focus on 'duration', while the latter focus on 'location'. For the data retention requirements, firms can retain data at any location, even abroad, whereas for local storage requirements data must be stored locally. Notably, the data retention and local storage requirements are often foundin different laws of a given economy.

endangers users' privacy. Hence, users may also be reluctant to engage with companies that will store their data for long periods, except in the case of certain types of data for whicha long retention period is necessary, such as medical records.

Data retention requirements can set out a **'minimum period of retention'**.[35] Under the 'minimum period of retention', firms must retain data for at least a specific period. The prescribed period can be days, months or years.[36] For example:

- In Bangladesh, data related to business transactions on e-commerce must be retained for 6 years from the transaction date.
- In Botswana, the Financial Intelligence Act requires that information obtained from the customers through customer due diligence, account files and correspondence should be retained for 20 years from the date the transaction was concluded and after the termination of the business relationship;
- In Botswana, Electronic Payments Services Regulations mandate that information should be retained for at least 5 years from the date that the transaction was concluded and after the termination of the business relationship;
- In India, cyber cafés are required to maintain a record of the user identification document, duly authenticated by both the user and authorized representative of the cybercafé for at least 1 year;
- In Nauru, service providers are required to store transmitted or host information for up to 12 months and to actively check for illegal activities to avoid liability;
- In Malawi, there is a required data retention period of at least 7 years;
- In Mexico, telecommunications concessionaires must keep the data of their users for at least 12 months in a system, and after this period, the data must be kept for an extra 12 months in an electronic storage system; and
- In Peru, concessionaires must keep specific telecommunications data related to the services provided for a period of 12 months in a computer system, and after this, the said data for an additional period of 24 months in an electronic storage system.

The score is '1' when the minimum period of data retention requirements is met. The score is '0' when there is no retention requirement or when a requirement exists but does not specify a minimum period, for example, a requirement to retain data as long as necessary.Additionally, Government data is not listed because the database focuses on commercial activities.

Data Protection Impact Assessment or Data Protection Officer requirements

This indicator asks whether an economy requires firms to appoint a Data Protection Officer (DPO) or perform a Data Protection Impact Assessment (DPIA). The DPOs ensure that a company processes personal data in compliance with data protection rules. The DPIA is a process of identifying risks of data processing operations on users' rights.[37] For example:

[35] There are two types of data retention requirements, 'minimum period of retention' and 'maximum period of retention'. The maximum period of retention is when firms cannot retain data longer than necessary without specifying a period. Businesses have latitude under this type of requirement in determining whether to retain data, because the limitations on the period during which businesses can draw value from personal data could be considered costly. However, without a clear period of retention requirement under the maximum period, it is not scored in the RDTII version 2.1.

[36] A somewhat atypical example of a minimum period of retention is a 'permanent period of retention.' Previously, India required the listed companies to permanently preserve the documents that are listed under Schedule I of the Securities and Exchange Board of India Regulations, such as incorporation documents, share certificates, register of minutes of board meetings and register of members. The score would be 1 for a permanent retention requirement. However, this requirement is no longer in effect.

[37] Notably, the GDPR mandates the DPIA for data processing activities "likely to result in a high risk." In the GDPR jurisdictions, the DPIA is commonly applied to the processing of sensitive data on a large scale, and a systematic and extensive personal aspect of an individual, including profiling (European Commission, 2018b).

- Colombia mandates that data controllers and processors must appoint a person, or division withinthe company, to assume responsibility for the protection of personal data, and be in charge of reviewing and solving claims made by data subjects;
- Ghana's Data Protection Act requires data controllers to appoint a data protection officer, also defined as a data protection supervisor, whose role is to monitor the data controller's compliance with the provisions of the Act;
- Singapore requires companies to appoint one or more data protection officers to ensure the organization's compliance with the Personal Data Protection Act. Organizations are also advised to conduct DPIA before releasing products or services that are likely to be accessed by children in order to meet the Accountability Obligation under the Personal Data Protection Act; and
- Türkiye requires companies to appoint "a data controller" who will be responsible for compliance under the Protection of Personal Data Law. If a controller is outside Türkiye, a local representative must be appointed. While DPO or a formal DPIA is not mandated, organizations registered in the Data Controllers Registry (VERBIS) must designate a contract person to communicate with the data protection authority. Data controllers are also required to take technical and administrative measures to ensure personal data security, including risk assessments.

While the purpose of these requirements is to ensure the data privacy of individual users and reinforce data protection, the measures may be costly. Firms, especially MSMEs, may struggle to hire a DPO with expertise in compliance with data laws across economies. Compounding this is that, unlike the European Union's General Data Protection Regulation (GDPR), data protection laws in developing economies often fragment. It is difficult for a DPO, even if appointed, to navigate through divergent data protection laws. A less costly alternative could be to make the appointment voluntary as an option to show companies' data privacy policies and practices.

Accordingly, the score is '1' if there is a requirement for the appointment of a DPOor the application of both DPO and DPIA horizontally across all sectors. The score is '0.5' if the requirements for having DPO or having both DPO and DPIA are applied only to a specific sector. The score is '0' when there is no requirement. Note that the requirement to perform the DPIA is generally introduced following the DPO appointment to ensure compliance with data protection rules. Hence, the scoring metric focuses more on the presence of the DPO requirement. The score of '0.25' would be applied if only DPIA is required. Such a case has not been found, but it is theoretically possible if a firm outsources the DPIA task to another company.

Requirements to allow government access to personal data

The indicator asks whether the Government can access personal data without the explicit authorization of an independent judicial body, such as a court decision,a judicial warrant or an equivalent order issued by a truly independent tribunal with due process safeguards. The lack of judicial oversight of this discretionary power could violate fundamental rights and thereby minimize trust in the digital environment. Government interference in one user's data may also create exploitable vulnerabilities in other accounts, such as:

- Cambodia mandates all telecommunications operators and related persons to provide telecommunications and ICT service data to the Ministry of Post and Telecommunications. Secret surveillance of telecommunications is allowed with the authorized of a legitimate authority. However, there is no definition of what constitutes a "legitimate authority," whether it is judicial authority, or the basis on which such authority is competent to approve surveillance;
- In Cuba, although exceptionally, the criminal investigator may use electronic surveillance as an investigative technique without the authorization of the Attorney General of the Republic;
- India requires ISPs to maintain a log of all users connected to their services. The ISPs must provide a complete list of subscribers with password-controlled access to the authorized intelligence agencies at any time; and
- Sri Lanka requires that all digital cellular mobile service operators are obligated to provide access to their databases or networks upon request by the Telecommunications Regulatory Commission for the purpose of obtaining subscriber information and data.

If this requirement applies, the score is '1'. The score is '0' when there is no requirement.

For this indicator, the World map of encryption laws and policies is also a useful secondary source. The secondary sources should only serve to guide researchers to the primary sources.

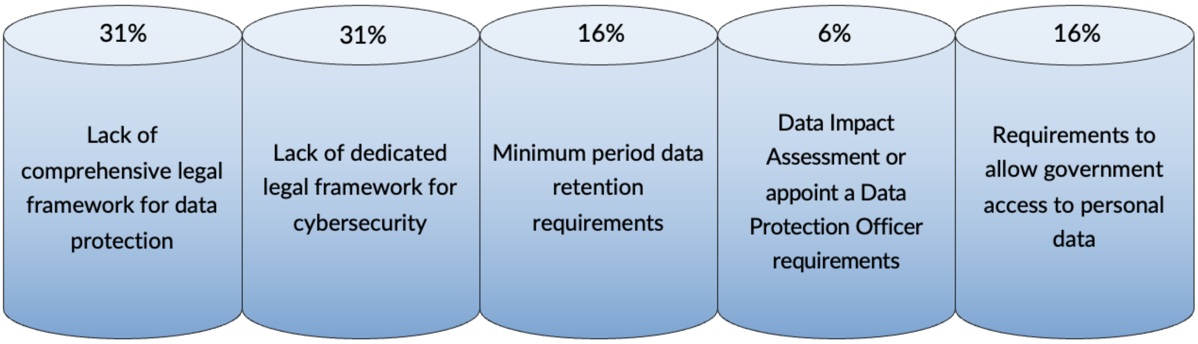
The weights for each indicator

As shown in figure 19, the weights for each indicator are 31%, 31%, 16%, 6% and 16%. The lack of a comprehensive data protection legal framework and a dedicated cybersecurity legal framework are given the greatest weight of 31% because having such a legal mechanism is crucial to promoting digital trust and legal certainty. A comprehensive data protection legal framework that applies across all sectors ensures users' data privacy and appurtenant rights and provides data protection mechanisms. Similarly, a dedicated cybersecurity legal ensures enforcement against cybercrimes. Conversely, the fragmentation of data protection and cybersecurity obligations that differ from sector to sector increases regulatory burdens on businesses.

The existence of a minimum period of data retention requirements and Government access to personal data are given the second greatest weight of 16% because both measures affect digital trust and discourage businesses. Although the second and the fourth indicators increase compliance costs and undermine data privacy, they are just one cybersecurity mechanism, whereas a comprehensive data protection law contains a bundle of users' rights and data protection mechanisms. Thus, its impact is more prevalent. In addition, the data retention requirements incur *ex ante* obligations on the part of businesses and create additional burdens. The Government's access to personal data for law enforcement purposes undercuts digital trust in the eyes of individual users.

The third indicator regarding the requirement to conduct DPIA or appoint a DPO is given the least weight (6%). This is because such a requirement negatively affects only one policy parameter – the cost of compliance; however, it still reinforces data privacy and data protection. The cost that it incurs, especially against MSMEs, does not justify the binding nature of the requirements.

Figure 19 **Pillar 7 indicators and the weights**

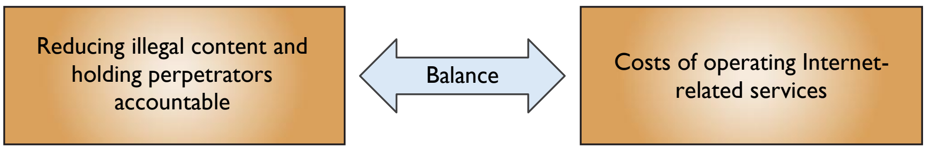


Pillar 8. Internet intermediary liability

Pillar 8 deals with measures governing intermediary liability. **'Internet intermediaries'** can be defined as "the intermediaries that bring together or facilitate transactions between third parties on the Internet.[38] They give access, transmit and index content, or provide Internet-basedservices to third parties." In this regard, they facilitate digital trade. The intermediaries include Internet service providers (ISPs) and Internet content providers (ICPs).[39]

Often, the regulation of Internet intermediaries is based on the interest of reducing illegal content over the Internet and holding perpetrators accountable. However, imposing onerous liability or certain obligations on the Internet intermediaries would likely discourage them from participating in digital trade in the regulated economy. Pillar 8 deals with regulatory measures that stand on an ill-placed balance of these two competing objectives, as shown in figure 20.

Figure 20 **Balance in different policy objectives in Pillar 8**



With the balance between these policy objectives, Pillar 8 consists of the following indicators:

- Lack of safe harbour for copyright infringements;
- Lack of safe harbour for other illegal activities;
- User identity requirements; and
- Monitoring requirements.

Useful secondary sources for this Pillar include the reports by the Global Network Initiative and the World Intermediary Liability Map. The secondary sources should only serve to guide researchers to the primary sources.

Lack of safe harbour for copyright infringements

This indicator asks whether an economy has a safe harbour provision for copyright infringement. A safe harbour provision protects the Internet intermediaries from legal liability for certain activities performed by their users. Without this provision, the Internet intermediaries hosting content in violation of copyrights will be, *per se,* legally responsible for their users' activities even when the intermediaries do not notice it. This may discourage investment in digital platforms. Furthermore, the safe harbour regime supports the emergence of innovative services as it provides intermediaries with legal certainty to conduct a wide range of activities, with less threat of potential liability and a less chilling effect of potential litigation.

[38] There is no agreed-upon definition of intermediaries. For more information, see the Oxford Handbook of Online Intermediary Liability (Frosio, 2020); see also OECD, 2011.
[39] Similarly, there is no agreed definition for ISPs and Internet content providers ICPs. ISPs can be defined as entities that provide Internet access and associated services such as email service, browser service, domain name registration and web hosting. There are several types of ISPs – for example, dial-up, cable (broadband), DSL (digital subscriber line) and fibre optics (satellite). ICPs can be defined as entities that disseminate online content to the end-users, such as social media platforms and news providers.

This provision that protects the intermediaries from legal liability for copyright-infringing materials can take various forms under sector-specific laws (e.g., for telecommunications or e-commerce service providers)[40] or be applied broadly across all sectors:

- The intermediaries are protected from legal liability for copyright-infringing materials in their platforms, and illegal activities are forbidden by different laws unless the intermediaries contributed to them or had notice of them beforehand. For example, for safe harbour provisions for copyright infringements, New Zealand's Copyright (New Technologies) Amendment Act protects ISPs from copyright-infringing content created by their users, if they do not have specific knowledge of the infringement or immediately make such content unavailable upon notice;
- The intermediaries are protected from legal liability for copyright-infringing materials in their platforms as well as illegal activities under the same law. For example, Japan'sAct on Measures Against Rights Infringement, etc. Arising from Distribution of Information by Specified Telecommunications (Act No.137 of 2001) protects telecommunications service providers from liability for damages caused by the distribution of information by specified telecommunications to the right of others unless it is aware of or has a good reason to be aware of the infringement. The telecommunications service providers also protect from any damage caused by the deletion of content on their networks if they reasonably believe that the content infringes intellectual property or the privacy of others, or if they sent a notice to the users of such content andhave not received a response within seven days; and
- The intermediaries are protected from legal liability for copyright-infringing activities but not for illegal activities. For example, the Republic of Korea's Copyright Act protects online service providers (ISPs and online hosts) from copyright infringement if it is technically impossible for them to take measures as prescribed in the listed requirements. Still, there is no safe harbour regime for other activities apart from copyright infringement.

The score is '1' for lack of a safe harbour provision for copyright infringement. The score is '0.5' if a safe harbour provision exists under a sectoral framework that limits intermediary liability for copyright-infringing activities. The score is '0' if a safe harbour provision exists under a horizontal framework that fully protects them from liability for copyright-infringing activities.

Lack of safe harbour for other illegal activities

This indicator asks whether an economy has a safe harbour provision for illegal activities other than copyright infringement. As mentioned, a safe harbour provision protects the Internet intermediaries from legal liability for certain activities performed by their users. Without this provision, Internet intermediaries hosting illegal content other than copyright infringement will be, *per se,* legally responsible for their users' activities even when the intermediaries are not noticed of it. This may discourage investment in digital platforms. Furthermore, the safe harbour regime supports the emergence of innovative services as it provides intermediaries with legal certainty to conduct a wide range of activities, with less threat of potential liability and a less chilling effect of potential litigation.

[40] For example, Georgia's Law on E-commerce regulates legal relations related to electronic commerce and protects intermediary service providers from liability for handling, caching, or hosting information under certain conditions. Specifically, they are exempt from responsibility if they did not initiate or alter the information, have no actual knowledge of illegal content, and act promptly to remove any illegal information upon becoming aware. Although the law does not explicitly mention copyright, it remains applicable.

The provision that protects the intermediaries from legal liability for activities apart from the copyright-infringing materials can take various forms under sector-specific laws[41] (e.g., for telecommunications or e-commerce service providers) or be applied broadly across all sectors:

- The intermediaries are protected from legal liability for copyright-infringing materials on their platforms and from illegal activities forbidden by different laws, unless the intermediaries contributed to them or had notice of them beforehand. For example, for other illegal activities apart from copyright infringements, New Zealand's Harmful Digital Communication Act provides protection to an online content host for illegal content posted by its users; and
- The intermediaries are protected from legal liability for copyright-infringing materials on their platforms and from illegal activities under the same law. For example, Japan's Act on Measures Against Rights Infringement, etc. Arising from Distribution of Information by Specified Telecommunications (Act No.137 of 2001) protects telecommunications service providers from liability for damages caused by the distribution of information by specified telecommunications to the right of others unless it is aware of or has a good reason to be aware of the infringement. The telecommunications service providers also protect from any damages caused by the deletion of content on their networks if they reasonably believe that the content infringes intellectual property or the privacy of others, or if they sent a notice to the users of such content and have not received a response within seven days.

The score is '1' for lack of a safe-harbour provision for activities other than copyright infringement. The score is '0.5' if a safe harbour provision exists under a sectoral framework that limits intermediary liability for activities other than copyright infringement. The score is '0' for a safe harbour provision that exists under a horizontal framework that fully protects them from liability for illegal activities other than copyright-infringing.

User identity requirements

The indicator asks whether an economy imposes user identity requirements. The **'user identity requirement'** mandates that Internet intermediaries verify and record accurate personalinformation of their users as a condition for accessing their services or networks. This measure includes collecting, storing and sometimes authenticating identifying documents or other identifying data. Such a measure can be costly in that the intermediariesare obliged to act as "gatekeepers" of the Internet, policing their users on behalf of the Government. This requires substantial efforts for the intermediaries to ensure that their users supply accurate personal information. For example:

- Argentina states that mobile communications service providers must store and systematise users' personalinformation in a registry of mobile service subscribers. Information that must be provided includes the name, surname, national identity document and address of users;
- Bangladesh requires that social media intermediaries providing messaging services enable the identification of the first originator of information and retain user registration data for 180 days after cancellation;

[41] For example, Georgia's Law on E-commerce regulates legal relations related to electronic commerce and protects intermediary service providers from liability for handling, caching, or hosting information under certain conditions. Specifically, they are exempt from responsibility if they did not initiate or alter the information, have no actual knowledge of illegal content, and act promptly to remove any illegal information upon becoming aware.

- Rwanda requires electronic communications service providers to ensure that their users supply accurate personal information when using a service or a network according to the Law Governing Information Communication and Technologies;
- Tajikistan requires users to present identity documents, such as passports or registration certificates, when concluding contracts for electronic communication services. Operators and providers must verify these documents before offering services.
- Uganda's Regulation of Interception of Communications Act requires intermediaries to collect customer information (name, address, identification number), install surveillance equipment and disclose information to the authorities upon the presentation of a warrant or a demand from the Minister for Information and Communications Technology and National Guidance; and
- In Venezuela, operators that provide mobile telephony services (prepaid or post-paid) must require their subscribers to provide personal information when contracting the referred service – for instance, their personal identification number, address, signature and fingerprints;

The score is '1' if there is a user identity requirement and is implemented in order to connect tothe Internet or access online services. The score is '0.5' if the user identity requirement is implemented for Subscriber Identity Module (SIM) card registration. Otherwise, the score is '0'.

Monitoring requirements

The entry asks whether an economy imposes content monitoring requirements. The **'monitoring requirement'** mandates that Internet intermediaries actively monitor the users' activities or remove or block content that is deemed illegal to avoid legal liability due to such content. Similar to theuser identity requirement, intermediaries are obliged to act as "gatekeepers" of the Internet, policing their content on behalf of the Government. Monitoring requirements can be explicit or indirect, such as requiring intermediaries to install software, algorithms, or other technological tools to detect or manage prohibited content. This responsibility demands substantial efforts for the intermediaries to monitor anything that is posted, shared or transferred by the users through the platform, and thereby could incur the cost. For example:

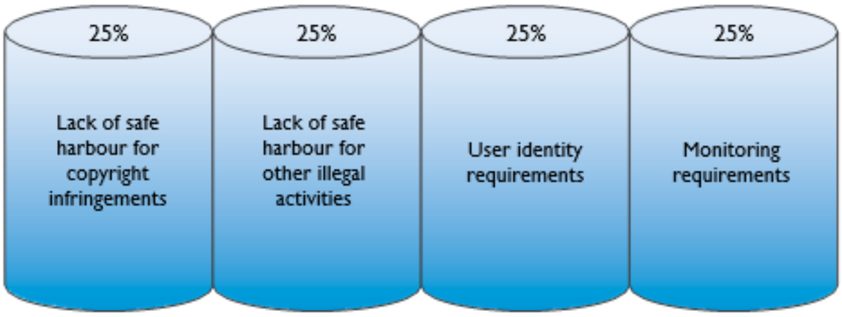
- Georgia requires internet site owners to periodically review links on their sites to ensure that the linked content does not contain offensive or inappropriate material, and to take appropriate measures to remove such content;
- Kazakhstan requires online platform providers to enhance content moderation systems and artificial intelligence algorithms to counter the placement and distribution of illegal content. The providers shall regularly report on moderation activities and user complaints;
- Lao PDR requires ISPs to be authorized to inspect the activities of their users and warn, stop or impose discipline. The website owners or website managers should also check their content thoroughly before allowing othersto disseminate the content through their websites; and
- Nepal mandates social network platform operators to verify the facts of content published or broadcasted on the platforms.

The score is '1' if at least one monitoring requirement is implemented. The score is '0.5' for the requirement of active monitoring of users' activities without any legal obligation to remove or block the content. Otherwise, the score is '0'.

The weights for each indicator

As shown in figure 21, an equal weight of 25% is given because each indicator seems to be an equally important requirement for Internet intermediaries. Regarding the first and second indicators, without the safe harbour provision in place, the intermediaries face a higher risk of being liable for their users' activities and the costs of litigation. The third and fourth indicators on user identity and monitoring requirements impose additional responsibilities on the intermediaries and thus incur costs. The intermediaries must procure additional software or hire additional workforce to monitor online activities and collect their users' identities. In addition,to mitigate the liability arising from their users' activity due to the lack of a safe harbour and the existing monitoring requirements, the intermediaries may impose unnecessary blocking and filtering, which are the measures under Pillar 9.

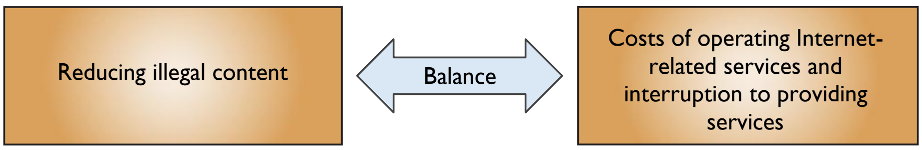
Figure 21 Pillar 8 indicators and the weights



Pillar 9. Content access

Pillar 9 deals with requirements on content access, which is somewhat connected to Pillar 8, Internet intermediary liabilities. Regulations of content access are based on the interest of reducing illegal content on the Internet. However, heavily regulated content access measures would increase the costs for intermediaries to operate Internet-related services and the costs of service interruption. Pillar 9 deals with regulatory measures that stand on an ill-placed balance of these two competing objectives, as shown in figure 22.

Figure 22 **Balance in different policy objectives in Pillar 9**



With the balance between these policy objectives, Pillar 9 consists of the following indicators:

- Blocking or filtering commercial web content;
- Internet shutdowns;
- Online advertising requirements; and
- Licensing requirements for online service providers.

Blocking or filtering commercial web content

This indicator asks whether there have been any instances of blocking or filtering commercial web content either by a Government or Internet intermediaries as required by the Government:

- **‘Blocking’** means denying access to a certain commercial website in its entirety;
- **‘Filtering’** is limiting access only to certain online content on a given website.

Generally, a Government blocks or filters websites or web content on the grounds of ‘public morality’ or ‘national security’ in order to prevent or respond to online security threats, such as malicious network traffic. Limiting content access of commercial websites or web content based onbroadly defined public policy grounds without clear and/or objective criteria constitutes a substantial burden on businesses and onlineusers because of the uncertainty of regulation and additional costs in managing their websites. For example:

- Brunei Darussalam bans or requires licensed Internet service providers and online content providers to use “their best efforts” to ban online content that is against the public interest or national harmony; and
- Türkiye empowers the authorities to block or remove online content for reasons including national security, public order, and crime prevention. Social network providers with over one million daily users must remove or block illegal content within 24 hours of notification and respond to user complaints within 48 hours. Failure to comply can result in administrative fines, advertising bans, or bandwidth throttling.

The score is ‘1’ for each blocking measure. All types and techniques of blocking, i.e., IP address and Protocol-based blocking, Deep Packet Inspection-based blocking, URL-based blocking and DNS-based blocking, are included because each type and technique leads to different outcomes, including under-blocking and over-blocking (Keller, 2018).[42] The score is ‘0.5’ for each filtering measure. Blocking or filtering political content, criminal content (e.g., child pornography), age-restricted content, defamation, and other non-commercial content would not be scored. Blocking or filtering based on intellectual property violations, such as copyright infringement content, is not considered a restriction because this content exploits the right holders’ exclusive right and is prohibited by the law; thereby, this would not be included, either.

Useful secondary sources include the reports by the Global Network Initiative, the World Intermediary Liability Map, and the Freedom House on the Net. The secondary sources should only serve to guide researchers to the primary sources.

Internet shutdowns

This indicator asks whether an economy imposes Internet shutdowns and the presence of Internet shutdowns occurring in the economy. Internet shutdowns refer to an international disruptionof the Internet or online services to prevent access to information, services and products. The shutdowns can be implemented in specific parts or areas within the economy. By implementing this measure, it has a large impact on the possibilities of doing business as well as affecting the users.

For the scoring of this indicator,

- The score is ‘1’ when Internet shutdown occurs extremely often. It is a regular practice for a Government to shut down domestic access to the Internet;
- The score is ‘0.75’ when Internet shutdown occurs often. The particular Government shut down domestic access to the Internet numerous times this year;
- The score is ‘0.5’ when an Internet shutdown occurs sometimes. The particular Government shut down domestic access to the Internet several times this year;
- The score is ‘0.25’ when Internet shutdown rarely occurs. The particular Government shut down domestic access to the Internet on a few occasions throughout the year;
- The score is ‘0’ when the Internet shutdown is never or almost never. The particular Government does not typically interfere with domestic access to the internet.

The mainsource is Varieties of Democracy (V-Dem) variable v2smgovshut (Annex II).[43] The V-Dem providestime series data in numerical form from 0 to 4. Under V-Dem, the interpretation of the score isdifferent from the RDTII 2.1, as follows:

- The V-Dem score of ‘0’ refers to when Internet shutdown occurs extremely often;
- The V-Dem score of ‘1’ refers to when Internet shutdown occurs often;
- The V-Dem score of ‘2’ refers to when an Internet shutdown occurs sometimes;
- The V-Dem score of ‘3’ refers to when Internet shutdown rarely occurs; and
- The V-Dem score of ‘4’ refers to when the Internet shutdown is never or almost never.

[42] For the clarification of each blocking technique, see https://www.internetsociety.org/resources/doc/2017/internet-content-blocking/
[43] V-Dem Codebook V.15 (as of March 2025) is available at https://www.v-dem.net/documents/55/codebook.pdf.

Online advertising requirements

The indicator covers requirements for online advertising. The measures could be limitations affecting online advertising, but exclude general consumer protection measures, such as prohibitions on misleading or false advertising, and restrictions on specific products and services (e.g., weapons, drugs, tobacco, alcohol, or medicine) for public health or safety reasons. The regulation that is silent about the scope of the application of requirements on advertisements can be considered as also being applied to online advertising, unless it is clear that the rule applies only to offline advertisements.[44] For example:

- The Marshall Islands mandates that advertisements must be published in both Marshallese and English, with Marshallese appearing first. Such advertisements must acquire approval from the Customary Law and Language Commission;
- Kazakhstan prohibits advertisements that contain a comparison of the advertised goods (work and service) with the goods (work and service) of other individuals or legal entities, as well as statements, images, discrediting their honour, and digital and business reputation; and
- The Russian Federation requires online advertising distributors, ad placement service owners, and aggregators of goods and services to allocate at least 5% of their annual advertising volume to unpaid social advertising, as well as report daily user access and advertising volumes.
- Uzbekistan prohibits advertisements that show prices in foreign currency.

The score is ‘1’ for each requirement limiting online advertising. Otherwise, the score is ‘0’. Any measure related to online advertisements for the purpose of consumer protection should also be scored ‘0’.

The secondary sources, such as the [OECD Digital STRI] should serve as guidance for the primary sources.

Licensing requirements for online service providers

This concerns a licensing scheme applicable to certain online service providers. In this context, online service providers refer to the following ICPs and applications:

- Social media platforms;
- News providers (e.g., media and broadcast services);
- Virtual Private Network (VPN); and
- Cloud services, etc.

This indicator excludes telecommunication facilities and service providers, which are already covered by Pillar 5, as well as e-commerce platforms, which are covered by Pillar 12. It asks whether (a) the ICPs are subjected to licensing requirements to operate their business, and (b) whether the licensing requirement is considered as **a “strict” requirement, in that they havethe potential to block businesses from operating their services.**

[44] The majority of countries have advertisement regulations that do not differentiate online and offline, and hence restriction is assumed to be applied to both.

A strict licensing scheme refers to a regime where the conditions for obtaining a license include requirements that significantly limit market access or competition. Such conditions go beyond normal regulatory standards and may involve requirements, such as commercial presence requirements, nationality or residency requirements, excessively high minimum paid-up capital, and technical or infrastructure requirements that mandate the use of government-approved systems, local data storage, or proprietary technology, which may be costly or restrictive.

Examples of strict licensing schemes:

- Indonesia requires Internet of Things (IoT) service providers to obtain a license or partner with a licensed telecom operator. The IoT service providers must implement a unique addressing system, including but not limited to a local Mobile Station International Subscriber Directory Number (MSISDN), device end-user ID, or IP number;
- Pakistan requires a license for broadcasting media and distribution services. However, such a license is prohibited to a person whois not a citizen or a resident in Pakistan, a foreign company, a company with the major shares owned or controlled by foreign natural or juristic persons, and any person funded or sponsored by a foreign Government or organization; and
- Viet Nam requires an aggregate of sites and socialnetworks to be operated by companies legally established in Viet Nam and meeting technical and operational standards in order to register for a license from the Ministry of Information and Communication.

The score is ‘1’ for a strict licensing scheme. The score is also ‘1’ if the economy implements more than one licensing requirement (even if not individually 'strict'). The score is ‘0.5’ if the economy implements just one licensing scheme. Otherwise, the score is ‘0’.

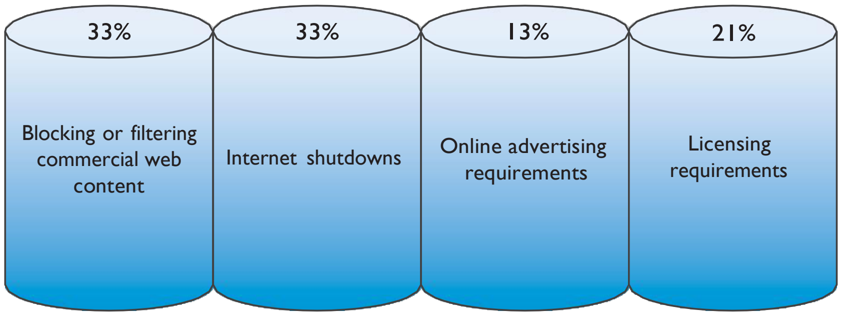
The secondary sources include the reports by the Global Network Initiative, the Freedom House on the Net, and the World Map of Encryption Laws and Policies. The secondary sources should only serve to guide researchers to the primary sources.

The weights for each indicator

As shown in figure 23, the weights of each indicator related to the Internet intermediaries are 33%, 33%, 13% and 21%, respectively. The first and the second indicators equally receive the highest weight of 33%. Blocking or filtering, and Internet shutdowns directly interrupt the intermediaries from performing their services, as well as having an impact on the end-users. The third and fourth indicators on the requirements for an online advertising and licensing scheme potentially discourage the Internet intermediaries without impeding their services, thereby receiving a lesser weight of 13% and 21%, respectively. Although online advertising requirements limit the intermediaries’ performance and could affect the engagement with customers, the intermediaries are still able to operate their services. Under the licensing requirements, the intermediaries may not be permitted to operate any activities unless they obtain a license. The license is also costly as it involves time and procedures.

Figure 23

Pillar 9 indicators and the weights

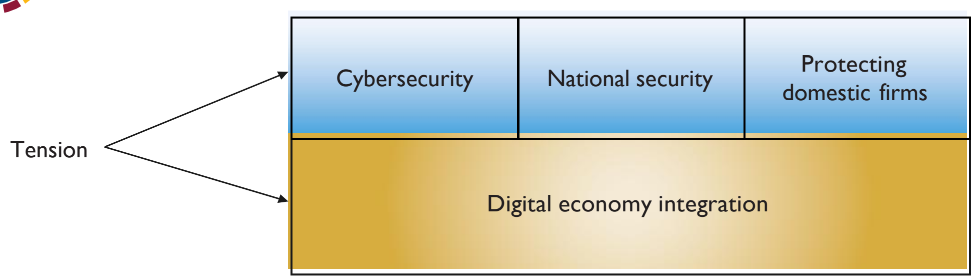


Pillar 10. Non-technical NTMs

Pillar 10 captures non-technical NTMs, including measures other than tariffs or taxes that limit the importation and exportation of ICT goods and online services from the economies within the considered United Nations region. These measures, such as bans, quotas and licensing procedures, could reduce the flow of ICT goods and online services.

Some of such measures are put in place based on the interests of protecting public order, cybersecurity and national security. For example, import bans on certain applications are designed to cut off cybersecurity as well as national security threats that they pose to the domestic networks. Certain export bans prohibit exports of certain ICT products and technologies thatthe economies consider to be vital to the interests of their nations. Other measures under this Pillar are for the purpose of guarding the domestic market against potential foreign market power.For example, local content requirements imposed on imports push foreign suppliers to acquire components that are to be built into their products from domestic suppliers. However, these policy objectives stand in tension, as shown in figure 24.

Figure 24 **Tension among different policy objectives in Pillar 10**



Pillar 10 aims to reveal specific areas of non-technical NTMs that this type of tension creates by covering the following indicators:

- Import bans;
- Other import restrictions;
- Local content requirements; and
- Export restrictions.

Import bans on ICT goods and online services

This indicator looks at import bans on ICT goods and online services. It covers bans that specifically target ICT goods and online services, as well as broader bans affecting multiple goods or services that include ICT products and online services. The score is ‘0.5’ if thereis only one ban covering a single specific product or service. A score of ‘1’ is assigned if there is more than one measure in place, or if a single measureapplies to more than one product or service. The score is ‘0’ if there is no such measure.

Secondary sources include the WTO I-TIP, the Global Trade Alert database, and the WTO Trade Policy Reviews. The secondary sources should only serve to guide researchers to the primary sources.

Other import restrictions on ICT goods and online services

This indicator covers import restrictions, excluding other than import bans and local content requirements on imports:

- Measures such as import quotas are trade-blocking, limiting the volume of ICT goods or online services that can be imported;
- Measures such as import licensing schemes and procedures do not necessarily block imports *per se*, but discourage the trade in ICT goods. They create additional costs and delay the import process. For example, in Argentina, non-automatic import licenses for ICT goods such as machines and apparatuses for the manufacture of semiconductors entail a more complex import process, in which additionaldocumentation and the intervention of technical agencies can take place. In Haiti, individuals of foreign nationality involved in importing are obliged to obtain a license, whichmust be renewed at the start of every fiscal year. In Botswana, the Communications Regulatory Authority (BOCRA) is mandatedto approve communications equipment that may be connected, used or operatedto provide broadcasting or telecommunications services in Botswana. Additionally, in Pakistan, only the Pakistan Television Corporation and other authorities licensed by the federal government are permitted to import transmission equipment for radio or TV, TV cameras, digital cameras, and video recorders.

The score is ‘0.5’ for each restriction that adds regulatory compliance costs to trade in ICT goods and online services, such as licenses, permits, authorization and registration for ICT goods, and labelling requirements and import controls, both for ICT goods and online services.[45] The score is ‘1’ for trade-blocking measures such as quotas, or if there are two or more measures that incur additional business costs, or the absence of institutional transparency. Institutional transparency implies that all public procurement laws, regulations, guidelines, and bidding procedures are publicly available, easily accessible (e.g., on a government website), and clearly communicate the requirements and decision-making processes. Otherwise, the score is ‘0’ if none of the above measures areimplemented.

Researchers should look at official laws, regulations, notifications, and other measures. The secondary sources are WTO I-TIP, the Global Trade Alert databases, the WTO Trade Policy Reviews as well as reports by international organizations. The secondary sources should only serve to guide researchers to the primary sources.

Local content requirements for the commercial market

This indicator covers **‘local content requirements’** (LCRs), i.e., requirements to use domestically manufactured goods or domestically supplied services in the production of ICT-related goods and online services. The LCRs under this Pillar do not include the measures implemented for public procurement tenders, which are covered by Pillar 2. For example:

[45] For clarification of each type of import-related procedure, see UNCTAD International Classification of Non-Tariff Measures, available at https://unctad.org/system/files/official-document/ditctab2019d5_en.pdf.

- Argentina has local content requirements for certain products (such as technical manuals, packaging and labels) in the production of mobile and cellular radio communication equipment operating in Tierra del Fuego province;
- Bhutan requires Over-the-Top (OTT) service providers to ensure that at least 60% of their content is local, promoting culturally and socially relevant material produced within the economy;
- India requires single-brand retail businesses, including e-commerce with physical stores, to source at least 30% of the value of their goods from India in order for foreign investors to own more than 51% of the company.
- Indonesia requires digital TV receivers and protocol set-top-boxes with a minimum local content requirement of 20% and a gradual increase to 50% within five years.

The score is ‘0.5’ when LCRs apply at the product level, i.e., HS-6 and HS-8 levels (e.g., mobile phones and smartphones).[46] The score is ‘1’ if there are two or more LCRs at the product level, or if the LCRs apply at the sectoral or horizontal level, i.e., HS-4 (e.g., telephony equipment) andHS-2 levels. The score is ‘0’ if there are no LCRs in this area.

Researchers should look at official laws, regulations, notifications and other measures of relevant ministries. Secondary sources such as WTO I-TIP, the Global Trade Alert database, the WTO Trade Policy Reviews as well as reports by international organizations should only serve to guide researchers to the primary sources.

Export restrictions on ICT goods and online services

This indicator covers export restrictions on ICT goods and services. These restrictions include export bans, export licenses and other restrictions limiting the number of goods or services to be exported. For example:

- Armenia, Georgia, India, Kyrgyzstan and the Republic of Korea require export licenses for strategic items, including dual-usegoods such as computers, telecom equipment, and information security products;
- Colombia prohibits the export of smartphones, with some exceptions such as mobile phones that are personal possessions of travelers, among others;
- Hong Kong, China, imposes export permits on radio transmitting apparatus; and
- Thailand requires an export license for radio communication equipment and ancillary devices.

The score is ‘1’ if at least one such measure is implemented. Otherwise, the score is ‘0’.

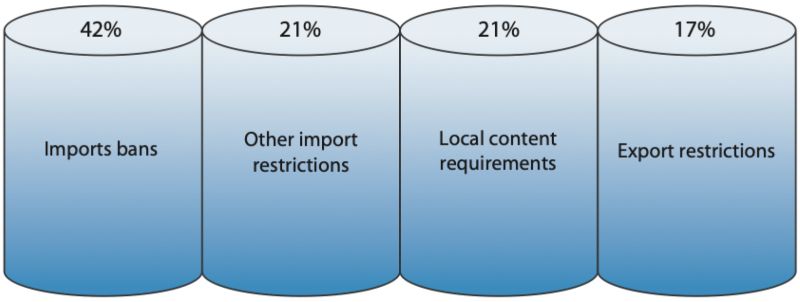
Researchers should look at official laws, regulations, notifications and other measures of relevant ministries. Secondary sources, such as WTO I-TIP, the Global Trade Alert database, the OECD Inventory on Export Restrictions on Industrial Raw Materials, the WTO Trade Policy Reviews as well as reports by international organizations should only serve to guide researchers to the primary sources.

[46] Harmonised System (HS) is an international nomenclature provided by the World Customs Organization (WCO) for the classification of products. The HS Code adopted a six-digit code system to classify goods (WCO, 2016).

The weights for each indicator

As shown in figure 25, the weights of each of the indicators are 42%, 21%, 21% and 17%. The first indicator on import bans gets the highest weight of 42% as it completely prohibits imports. Indicators on other import restrictions and local content requirements equally receive a lesser weight of 21% because these measures limit the flows of ICT goods and online services, as well as discriminate against foreign businesses. Other import restrictions hinder trade facilitation by imposing additional import procedures. Local content requirements require foreign companies to use domestic resources according to the prescribed threshold, increasing costs and barriers for the companies to find suitable local materials or services for their supply chains. Each of the measures directly undermines foreign competition in the domestic market, while the last indicator, on export restrictions, has a limited impact on foreign competition in the economy concerned and is thereby given the least weight of 17%.

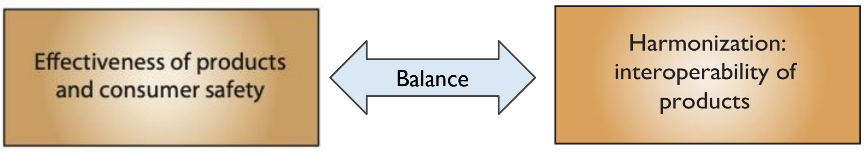
Figure 25 **Pillar 10 indicators and the weights**



Pillar 11. Standards and procedures

Pillar 11 covers technical standards and related procedures that can function as a trade restriction on ICT goods and online services in a digital economy, specifically in the telecommunication sector. Technical standards ensure the effectiveness of products and consumer safety by setting out the minimum requirements necessary to ensure the quality of products or services. However, disparate technical standards and related procedures across the region undermine this interoperability and thereby discourage digital trade due to higher trade costs and delayed processes. Adopting technical standards that are internationally recognized as best practices increases the interoperability of products and services across the region (figure 26).

Figure 26 **Balance in different policy objectives in Pillar 11**



Pillar 11 measures the interoperability of regulation in the following indicators:

- Technical standards issues;
- Self-certification limitations for product safety (radio transmissions, EMC/EMI);
- Product screening and testing requirements; and
- Deviation from international encryption standards.

Technical standards issues

In developing an open and transparent standards regime, all relevant stakeholders should be able to engage and provide comments. This indicator asks whether (a) participation from relevant stakeholders, including foreign businesses, is not allowed or restricted in the standard-setting of the standards in sectors relevant to digital trade, and (b) whether there are transparent public consultation or engagement mechanisms exist. Restrictions may include, for example:

- The limit on foreign participation. For example, in Cuba, foreign companies are not allowed to participate in institutional bodies that establish and regulate trade norms. Egypt requires that the board of directors of the National Telecommunication Regulatory Authority is authorized to set standards for telecom equipment without foreign participation; and

- The lack of institutional transparency. This can be found in various forms, such as the absence of clear consultation processes or the lack of public notification to invite stakeholder input.

This indicator has binary scores, ‘0’ or ‘1’. The score is ‘1’ if foreigners are not allowed to participate in the standard-setting bodies or if standard-setting is not transparent. Researchers should look at official laws, regulations and other measures. Secondary sources, such as the WTO Trade Policy Reviews, should only serve to guide researchers to the primary sources.

Self-certification limitations for product safety (radio transmissions, EMC/EMI)

This indicator asks whether the self-certification of product safety by suppliers is not allowed. Certification refers to a process of verifying that imported products meet domestic standards. For instance, electrical products generally must comply with domestic standards of radio transmissions, electromagnetic interference (EMI) or electromagnetic compatibility (EMC).[47]

Economies allow the self-certification of the products to a varying degree. Generally, economies implement the following measures for self-certification and third-party certification. These processes result in the issuance of a certificate or declaration of conformity for the imported products:

- Self-certification through **‘Supplier’s Declaration of Conformity’** (SDoC) ensures compliance with the prescribed domestic standards;
- Third-party certification from **‘Conformity Assessment Bodies’** (CABs) does away with the need for local testing of the products to be exported. Generally, economies that are members of a Mutual Recognition Agreement (MRA), such as the ASEAN MRA for Electrical and Electronic Equipment (ASEAN EE MRA) and APEC MRA for Conformity Assessment of Telecommunications Equipment (APEC TEL MRA), maintain reciprocity in recognizing CABs in their territories (ASEAN, 2012; FCC, 2016).

The score is ‘1’ if an economy recognizes neither self-certification nor third-party certification and requires foreign suppliers to undergo testing in a local laboratory. The score is also ‘1’ if an economy does not have any regulatory framework or indication for accepting SDoC. The score is ‘0.5’ if a SDoCis not permitted, but the third-party certification from CABs in other economies is accepted. The score is ‘0’ if an SDoC is permitted for foreign businesses.

It is important to note that an economy may allow SDoC for specific types of ICT products from foreign businesses while requiring third-party certification for others. In such cases, if there is a regulatory framework that recognizes the submission of SDoC for at least some ICT products, the assigned score is ‘0’ since it is not expected that an economy will allow SDoC for all types of ICT products due to varying levels of technical safety and complexity. Similarly, if an economy recognizes third-party certification for specific types of ICT products but requires local testing for others without accepting SDoC or third-party certification, the assigned score will be ‘0.5’.

Researchers should look at official laws, regulations and other measures. Secondary sources, such as the reports by business associations and from the WTO Trade Policy Reviews, the Telecommunication Industry Association (TIA) (which represents manufacturers and suppliers of global communication networks) and the International Telecommunication Union (ITU) should serve only to guide researchers to primary sources.

[47] The EMC testing measures whether electrical devices can function in the environment without interfering with surrounding equipment by emitting radiation. The EMI testing gauges whether electrical products can function in the presence of a certain amount of electromagnetic interference. Different requirements and interpretations of the definition of EMC and EMI in the United States and the European Union could cause confusion when it comes to testing (Hayes, 2021).

Product screening and testing requirements

The indicator asks whether an economy imposes an additional screening or testing requirement on ICT imports. Additional screening or testing refers to processes where products are required to undergo further technical tests beyond standard conformity assessments to verify certain aspects of compliance before they are allowed into the market. The declared justifications for these requirements are often about national security.[48] For example:

- The Gambian standards generally require local testing for electrical products for certification. Audio and video products such as televisions, LCD panels and similar apparatus marketed in The Gambia are required to undergo local testing. Audio and video products are certified only after conformity assessments have been carried out by The Gambia Standards Bureau (TGSB);
- India requires in-economy security testing on all telecom equipment before sale, import, or use. Testing must be conducted by those designated by the Telecom Engineering Center (TEC), and certificates are issued based on their test reports; and
- Sri Lanka requires importers of designated goods, such as primary cells, batteries, and PVC-insulated non-armoured cables, to submit documentation to Customs and the Sri Lanka Standards Institution (SLSI) before clearance. Product samples must undergo testing for conformity with Sri Lankan standards. These products cannot be distributed without prior approval from the Director General of the SLSI.

If a requirement for domestic screening or testing exists, the score is '1'. This includes cases where testing must be conducted by in-country designated or country-accredited assessment bodies. If third-party testing results are accepted by local authorities, the score is '0.5'. The score is '0' if there is no requirement for screening or testing.

It is important to note that an economy may impose additional screening or testing requirements for certain types of ICT products while accepting third-party testing results for others. In such cases, if a regulatory framework permits acceptance of third-party testing at least for some ICT products, the assigned score is '0.5', reflecting partial acceptance. However, if domestic screening or testing is required for certain products without acceptance of third-party testing, the assigned score is '1'.

Researchers should look at official laws, regulations, notifications, and other measures. Useful secondary sources include reportsby business associations and from the WTO Trade Policy Reviews, the Telecommunication Industry Association (TIA) and the International Telecommunication Union (ITU). The secondary sources should only serve to guideresearchers to the primary sources.

Deviation from international encryption standards

The indicator asks whether an economy adopts encryption standards that deviate from internationally recognized standards based on the principles in the International Organization for Standardization (ISO) and International Electrotechnical Commission (IEC) encryption standards.

[48] Thus, these requirements are distinguishable from testing or screening requirements based on the interest of public safety and efficiency of products, such as EMC or EMI testing requirements.

Encryption (or encipherment) is the process of encoding a message or information with an algorithm by converting original text (known as plaintext or cleartext) toan alternative form (known as ciphertext).[49] Decryption, in turn, is the reverse process of accessing the plain text of the encrypted message, requiring a password or a **'private key'**. The objective of encryption is to secure data and prevent data breaches. The encryption strength is based onthe key's length, block size and design.

To determine whether an economy's encryption requirements deviate from internationally recognized standards, the assessment is based on the ISO/IEC standards relevant to encryption, particularly ISO/IEC 18033 series (encryption systems),[50] and ISO/IEC 11770 series (key management).[51] Deviations may occur in the following cases, for example:

- **Algorithm.** The requirement to use encryption algorithms not listed in the encryption standards, such as the requirement to adopt domestic encryption algorithms. The ISO/IEC 18033 specifies the recognized algorithms, such as TDEA, MISTY1, CAST-128, HIGHT, AES, Camellia, and SEED;
- **Block size requirements.** The requirement to use block sizes below the ISO/IEC 18033 recommended minimums, i.e., lower than 64-bit block cyphers and 128-bit block cyphers;
- **Key length.** The requirement to use weaker keys than ISO/IEC 18033 recommended, which are 128 bits or more for symmetric encryption;
- **Key management.** The requirement to use key management is not listed in the ISO/IEC 11770. The criteria are broad, such as key methods for asymmetric key include RSA-based key transport, and Elliptic Curve Cryptography (ECC)-based key exchange; and
- **Disclosure of proprietary information.** The disclosure of trade secrets (e.g., source code, encryption keys) when certifying encryption products beyond the established measures under the ISO/IEC validation requirements (ISO 19790 and ISO 24759). ISO/IEC standards include testing and validation of cryptographic modules, but do not mandate trade secrets disclosure.

Note that the registration and licensing requirements in order to use, import, or distribute an encryption device are not considered as a restriction under this indicator if the conditions do not include issues mentioned above. Instead, such licensing requirements shall be covered under other pillars, such as pillar 9 and pillar 10.

The score is '1' if at least one of such measures is implemented. Otherwise, the score is '0'.

Researchers should look for official laws, regulations, notifications, and other measures. Secondary sources include the World Map of Encryption Laws and Policies and the reports of the Freedom House on the Net. The secondary sources should only serve to guide researchers to the primary sources.

[49] Encryption is a principal application of cryptography. Cryptography refers to the technological means to secure information and communications systems (OECD, 2022).

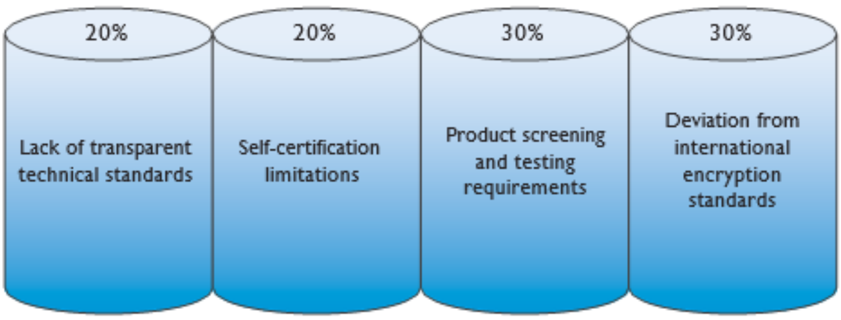
[50] The ISO/IEC 18033 series defines encryption systems (cyphers) designed to ensure data confidentiality. The series currently consists of seven parts, outlining standards for six types of encryption algorithms: asymmetric cyphers, block cyphers, stream cyphers, identity-based cyphers, homomorphic encryption, and tweakable block cyphers.

[51] The ISO/IEC 11770 series specifies methods for key management in cryptographic systems, including encryption and other cryptographic operations. It addresses key generation and management techniques for both symmetric and asymmetric cryptography. Symmetric cryptography uses a single key for both encryption and decryption, while asymmetric encryption algorithms refer to cryptographic methods that use two separate keys—a public key for encryption and a private key for decryption.

The weights for each indicator

As shown in figure 27, the weights are 20%, 20%, 30% and 30%, respectively. The third and fourth indicators are assigned similar weights of 30% because each indicator has the potentialto discourage foreign businesses from operating in certain economies. As new digital technologiesare increasingly emerging, standard-setting on ICT goods and online services is crucial for creatingan integration system and a timely response to technological developments. Divergent domestic approaches to standards and compliance procedures of self-certification and testing requirements could result in inconsistent quality and safety of the end-products or services in the market. The first and second indicators receive less weight compared with 30%. Although self-certification requirements and screening requirements provide additional layers and thus slow down the process, the product screening and testing requirements could further result in a banon the imported ICT goods or prohibition of certain online services. The screening and testing mechanisms are generally based on the reason of national security, and thereby could create uncertainty. The requirement on encryption standards and trade secrets not only leads to a lack of interoperability, as it does in the technical standards regime. The encryption in relation to data confidentiality of the data storage and data transmission against unauthorized access, as well as the mandatory disclosure of trade secrets, creates impacts on business competition.

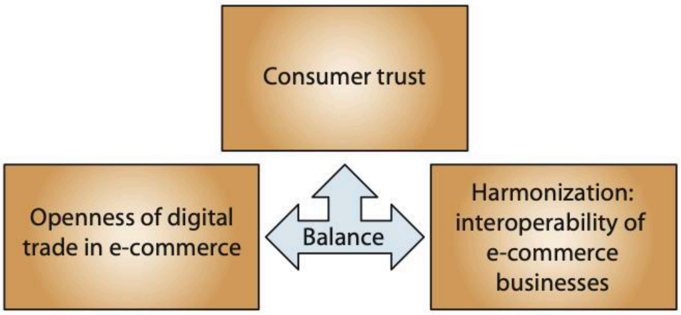
Figure 27 **Pillar 11 indicators and the weights**



Pillar 12. Online sales and transactions

Pillar 12 captures requirements for online sales and transactions. The steady increase in online sales and transactions over the years, both in developed and developing economies, reflects how critical these flows have become for digital trade. Measures surrounding electronic commerce in the areas of foreign investment, online sales, delivery, licensing requirements, online payment, *de minimis* rule, domain names and local presence requirements may limit digital trade as these are essentials to set up and operate an e-commerce business in the regulating economy. Furthermore, the lack of legal recognition for electronic communication, signatures and transactions cuts back digital trust and undermines the interoperability of e-commerce businesses in the region. Finally, the presence of consumer protection laws in the e-commerce sector increases consumer trust and, thereby, participation in digital trade by foreign vendors in the regulating economy. Sound policies for e-commerce find a balance among these policy objectives, as shown in figure 28.

Figure 28 **Balance in different policy objectives in Pillar 12**



Based on this understanding, Pillar 12 reflects the following issues:

- Foreign equity limits in the e-commerce sector;
- Online purchases and delivery limitations;
- Licensing requirements in the e-commerce sector;
- Online payment limitations;
- Low De Minimis;
- Imposition of customs duties on electronic transmission;
- Domain name requirements;
- Local presence requirements;
- Lack of a legal framework for online consumer protection;
- Not in the United Nations Convention on Electronic Communications;
- Not in the UNCITRAL Model Law on Electronic Commerce;
- Not in the UNCITRAL Model Law on Electronic Signatures; and
- Not in the UNCITRAL Model Law on Electronic Transferable Records.

Foreign equity limits in the e-commerce sector

This indicator concerns the maximum foreign equity shares in the specific sector of e-commerce. Foreign equity shares are shares that foreign natural or legal persons hold in a firm incorporated in the investee economy. Limitation on foreign equity shares is a direct obstacle for foreign investors that could induce a higher level of development in the e-commerce sector. For example, the Philippines bans foreign ownership of retail trade enterprises with paid-up capital of less than US$ 2.5 million.

In particular, the foreign equity shares under this Pillar do not include the foreign equity shares in other sectors relevant to digital trade or the telecommunication sector, which are captured in Pillars 3 and 5, respectively. According to a specific focus on e-commerce, the scoring metric on State-owned enterprises is not included.

The score is ‘1’ if only a minority stake (less than 50%) is allowed. The score is ‘0.5’ where a controlling stake (more than 50%) is allowed, but maximum caps on foreign equity exist. The score is ‘0’ if there is no limitation on foreign equity share in the e-commerce sector.

Online purchases and delivery limitations

The indicator covers direct requirements on online purchases and the requirements on the delivery of products bought online that affect consumers or retailers. The measures could be:

- Specific limits on the number of goods imported by customers through e-commerce platforms. For example, Brazilian Customs have established express services with maximum per-shipment value limits of US$ 3,000 for imports, while in Argentina, consumers can purchase goods valued at up to US$ 50 per month tax-free, with an annual tax-free limitof US$ 600. If the monthly purchase total exceeds US$ 50, the consumer must paya 50% tax on the value above the US$ 50 threshold;
- Specific limits on the number of goods to purchase through e-commerce platforms; and
- Limitations on the delivery of products bought online applied to the delivery company. For example, Indonesia prohibits foreign postal operators from operating intercity delivery services and requires them to cooperate with local delivery service companies for deliveries at the provincial level; and
- Limitations on the delivery of products bought online applied to the users, such as restrictions on how, where or when consumers can receive deliveries.

This indicator does not cover measures related to taxation, customs duties, or fees imposed on online purchases or deliveries. Limitations on online purchases and delivery that are applied specifically for consumer protection purposes, such as restrictions on alcoholic beverages, tobacco, and pharmaceuticals, are not considered as a restriction under this indicator. In addition, measures related to foreign direct investment (such as foreign equity caps, commercial presence requirements) targeting companies, such as delivery service providers, are excluded from this indicator and should instead be recorded under the relevant indicators in Pillar 3.

The score is ‘1’ if at least one requirement limits the number of purchases via e-commerce or delivery. Otherwise, the score is ‘0’.

Researchers should look at official laws, regulations, and other measures. Secondary sources, such as the OECD Digital STRI should serve as guidance for the primary sources.

Licensing requirements in the e-commerce sector

The indicator covers any licensing scheme for e-commerce providers, including both business-to-business (B2B) and business-to-consumer (B2C) platforms. Given the diverse licensing landscape within the e-commerce sector, this indicator specifically focuses on the license imposed on e-commerce businesses. Licenses pertaining to other aspects of e-commerce businesses, such as a license to operate online payment services and delivery, are not captured.

For example:

- Bhutan mandates that a company or a Bhutanese individual aged 18 or older must obtain a license to operate an e-commerce business. Prior approval from the Department of Trade is required for those wishing to operate a national e-commerce portal or platform;
- Colombia requires web pages and Internet sites of Colombian origin whose activity is of a commercial, financial or service nature to be registered in the commercial registry, and to provide the National Tax and Customs Directorate with information on their economic transactions;
- The Lao PDR requires e-commerce businesses, both B2B and B2C, that trade through their platforms or electronic marketplaces to obtain an acknowledgement certificate from the Ministry of Industry and Commerce. Electronic marketplaces must also be incorporated entities, obtain an e-commerce business license from the Ministry, and secure a technical conformity certificate from the Ministry of Technology and Communications; and
- Türkiye requires that e-commerce service providers with annual net transactions of more than 10 billion Turkish Lira and 100,000 transactions excluding cancellations and returns are required to obtain a license from the Ministry.

The score is ‘1’ if at least one licensing requirement for e-commerce providers. Otherwise, the score is ‘0’. Researchers should look at official laws, regulations, and other measures. Secondary sources, such as the OECD Digital STRI should serve as guidance for the primary sources.

Online payment limitations

The indicator covers requirements for online payments and other requirements affecting the use of electronic payment and credit services. Such requirements are diverse:

- Requirements to use a local bank account;
- Requirements on the currency used for international payments;
- National standards for payment security that deviate from international standards;[52]
- Licensing requirements with restrictive conditions (such as local incorporation requirements, nationality restrictions, or limitations on the types or numbers of licenses that can be obtained);
- Ceilings on the maximum amount that can be paid by electronic payment methods; and
- Requirements mandating the use of specific intermediaries for online payments.

[52] The indicator adopts a flexible approach. It recognizes any international standards referenced in the law related to online payment security, including those developed by the ISO/IEC, the PCI DSS (Payment Card Industry Security Standards Council), and other recognized frameworks.

For each measure, the score is '1' if *any* measure is present.[53] Otherwise, the score is '0'. Requirements for 'cryptocurrencies'[54] are not listed since the implication for using this type of payment is relatively new and still lacks concrete evidence. Researchers should look at official laws, regulations and other measures. Secondary sources, such as the OECD Digital STRI, should serve as guidance for the primary sources.

Low de minimis

The indicator asks whether an economy adopts a threshold for the *de minimis* rule. The *de minimis* rule sets a valuation ceiling for goods below which no duty or tax is charged at the border to ensure the flow of digital trade. According to the International Chamber of Commerce (ICC) recommendation of establishing a global baseline of *de minimis* value, the index calculates the valuation ceiling based on a US$ 200 threshold (ICC, 2016).

If no *de minimis* rule exists, a score of '1' is given. If an economy adopts the *de minimis* rule below US$ 200, a score of '0.5' is given. However, if the *de minimis* rule is equal to or above US$ 200, a score of '0' is given. For consistency across datasets, the exchange rates should be based on the International Monetary Fund (IMF). Researchers should look for relevant official laws, regulations, notifications and other measures. A useful secondary source is the Global Express Association database. The secondary sources should only serve to guide researchers to the primary sources.

Imposition of customs duties on electronic transmission

This indicator asks whether an economy imposes customs duties or cross-border duties on electronic transmission. The electronic transmission could refer to the import and export of digital products by electronic means.[55]

Legal mechanisms or regulations allowing for the imposition of such duties, even if current tariff lines are zero, will be considered. For example, in Indonesia, no customs duties are currently imposed on electronic transmissions. However, the economy has established regulations, which classify intangible goods which are delivered via electronic transmission, such as software, electronic data, and multimedia, under specific HS codes (Heading 99.01) with a 0% duty rate, indicating a legal framework for such duties exists.

For scoring, if an economy does not implement any legal mechanism or regulation that is applicable to impose customs duties on electronic transmission, the score will be '0'. If an economy has legal mechanisms or regulations allowing for the imposition of such duties, even if current tariff lines are zero, the assigned score is '0.5'. The score of '1' will be assigned if the economy has actively imposed customs duties on electronic transmissions in practice.

[53] While the score of the indicator is not a cumulative sum of these individual measures, a cumulative weight is applied to reflect the breakdown of the indicator.
[54] Cryptocurrencies, a subset of virtual currencies, adopt blockchain or distributed ledger transactions (DLT) to process virtual transactions. The entire concept of digital currency is available at (ESCAP, 2017); see also Houben and Snyers, 2018.
[55] Although the electronic transmission scope, definition and the impact of the Moratorium has been negotiated at the WTO Moratorium on Custom Duties on Electronic Transmission since 1998, the term electronic transmission itself is not explicitly or formally defined under WTO law. At the 13th Ministerial Conference 13 (MC 13), members agreed to extend the moratorium for another two years. For more information, see https://www.wto.org/english/tratop_e/ecom_e/ecom_work_programme_e.htm

Domain name requirements

The indicator covers requirements on commercial domain names.[56] These requirements may have costs of compliance. They include requirements for companies to have a local domain name to engage in electronic retail in a certain market, requirements to establish a local presence as a condition for using a local domain name, and requirements to appoint a local representative.

For example:

- In Brazil, foreign companies aiming at registering a domain must have a representative in the country to register before the country's official registry administrator for domain names;
- New Caledonia restricts the registration of ".nc" domains to individuals and legal entities established in the territory. Individuals must have resided in New Caledonia for at least 6 consecutive months, while legal entities must have a registered office or branch locally. Moreover, all applicants must appoint an administrative contact based in New Caledonia who can legally receive judicial or extrajudicial notifications; and
- Viet Nam requires licensed online newspapers, websites, web portals and social networking sites to use the ".vn" domain name.

All commercial country-code top-level domains (ccTLDs) and their sub-domains are included. However, the domain names for government agencies, military, educational institutions or other organizations, namely '.gov', '.mil', '.edu' or '.org', are not listed because they do not focus on commercial activities.

A score of '1' is assigned if a physical presence is required to use a local domain name or if companies are required to obtain a local domain name in order to engage in electronic commerce business. A score of '0.5' is given if companies are mandated to appoint a local administrator.If there are no requirements on the domain name, the score is '0'.

Local presence requirements

This indicator asks whether an economy imposes local presence requirements on online service providers. The **'local presence requirement'** requires companies to have a physical representative office, local agent, legal representative or a post-box within the economy to provide their services.

This indicator focuses on the requirements that serve as a precondition for foreign companies to provide online services across borders (Mode 1), without ever establishing a physical office.

This requirement differs from the commercial presence requirement (Mode 3). While the commercial presence involves establishing a local subsidiary or branch, the local presence requirement only mandates administrative representation (e.g., an agent or contact point) when foreign providers deliver services digitally or remotely.[57]

It is important to note that requirements to maintain a registered office or registered agent, typically arising under corporate law, they are not considered local presence requirements recorded under this indicator, as they apply only when a company has already established a legal

[56] A domain name system (DNS) links the online users to each IP address, which is a string of numerical digits and periods, by providing a familiar string of letters known as the 'domain name'. A domain name includes different types and hierarchies: Top-Level Domains (TLDs), Second-level domains and Third-level domains. It refers to .com, .net, .org within a specific ccTLD (e.g., .vn, .br) that are used for commercial purposes.
[57] For more information, see the definition of services trade and mode of supply available at https://www.wto.org/english/tratop_e/serv_e/cbt_course_e/c1s3p1_e.htm.

presence (Mode 3). This means the requirements are not preconditions for cross-border online service provisions (Mode 1). In other words, this indicator **only** counts rules that require a foreign online service provider to have a local contact point as a condition for providing services remotely, **even without having a physical office**.

When both local presence and commercial presence requirements are imposed under the same law (e.g., a law requires a foreign provider to register as a legal entity and appoint a local representative), the measure should be classified as a commercial presence requirement because the establishment of a local entity is the stronger obligation. The appointment of a representative is inherent once a company establishes itself locally.

For example:

- Indonesia requires Private Electronic System Operators (ESOs) to register their businesses at the relevant ministry. The ESOs should appoint liaison officers who have a domicile in Indonesia to facilitate any access request by the government authorities;
- The Republic of Korea requires foreign IT service providers without a physical office in the economy to designate a local agent responsible for data privacy compliance; and
- Türkiyerequires foreign-based social network providers whose platforms are accessed from within Türkiye more than one million times a day to appoint a real or legal person representative locally.

For scoring, if there is a local presence requirement, the score is ‘1’. Otherwise, the score is ‘0’. Researchers should look at official laws, regulations, and other measures. The secondary source, which is the DSTRI database should only serve to guide researchers to the primary sources.

Lack of legal framework for online consumer protection

This indicator asks whether an economy has adopted an online consumer protection legal framework. The domestic laws of consumer protection need not be specific to e-commerce. A law that applies to transactions across sectors for the purpose of protecting consumers can extend to e-commerce transactions under its umbrella. The consumer protection law applicable to online purchases can be found within the same protection provided to offline transactionsor in a separate regulation.

The score is ‘1’ if an economy lacks an online consumer protection legal framework or if the economylacks consumer protection laws that are applicable to online purchases. The score is ‘0’ for the presence of online consumer protection laws.

Researchers should look for official laws and regulations. UNCTAD Cyber Tracker database is a useful secondary source that can guide researchers’ attention to the primary sources.

Not in the United Nations Convention on Electronic Communications

The indicator asks whether an economy has signed and ratified the United Nations Convention on the Use of Electronic Communications in International Contracts (2005) (the Electronic Communications Convention). The Electronic Communications Convention is a binding treaty which requires member economies to recognize the legal validity and enforceability of electronically concluded contracts and other communications exchanged electronically (UNCITRAL, 2018c) (box 6).

If the economy has not signed and ratified the mentioned international framework, the score is ‘1’. If the economy has signed but not ratified the framework, then the score is also ‘1’. If an economy has signed and ratified the Convention, the score is ‘0’.

To see the status of the Convention, check the United Nations Convention on the Use of Electronic Communications in International Contracts (status) or the official government website.

Not in the UNCITRAL Model Law on Electronic Commerce

The indicator asks whether an economy has adopted the UNCITRAL Model Law on Electronic Commerce (1996) (MLEC). The MLEC is a model law on which an economy may base its legislation fully or in part. The MLEC establishes rules for the formation and validity of contracts concluded by electronic means, attribution of data messages, acknowledgement of receipt, and determining the time and place of dispatch and receipt of data messages (UNCITRAL, 2018a) (box 6).

If the economy has neither adopted any parts of the mentioned international legal framework nor notified the adoption of the UNCITRAL Model Law on the UNCITRAL Secretariat, the score is ‘1’. If the economy has adopted fully or in part the Model Law,[58] but has not notified the UNCITRAL Secretariat, the score is ‘0.5’. If the economy has adopted fully or in part the Model Law and notified the UNICITRAL Model Law, the score assigned is ‘0’.

To see the status of the model law, check the UNCITRAL Model Law on Electronic Commerce (status) or the official government website.

Not in the UNCITRAL Model Law on Electronic Signatures

The indicator asks whether an economy has adopted the UNCITRAL Model Law on Electronic Signature (2001) (MLES). Similar to the MLEC, the MLES is a model law on which an economy may base its legislation fully or in part. The MLES establishes criteria of technical reliability for the equivalence between electronic and hand-written signatures as well as basic rules of conduct that may serve as guidelines for assessing duties and liabilities for the signatory, the relying party and trusted third parties intervening in the signature process (UNCITRAL, 2018b) (box 6).

If the economy has neither adopted any parts of the mentioned international legal framework nor notified the adoption of the UNCITRAL Model Law on the UNCITRAL Secretariat, the score is ‘1’. If the economy has adopted fully or in part the Model Law,[59] but has not notified the UNCITRAL Secretariat, the score is ‘0.5’. If the economy has adopted fully or in part the Model Law and notified the UNICITRAL Model Law, the score assigned is ‘0’.

To see the status of the Model Law, check the UNCITRAL Model Law on Electronic Signatures (status) or the official government website.

[58] Adoption of a UNCITRAL Model Law in whole refers to national legislation that closely follows the full text or core provisions of the Model Law. Partial adoption means that only some provisions are incorporated, or the law is broadly influenced by the Model Law’s principles. The notes (a) to (h) on the UNCITRAL Model Law on Electronic Commerce status webpage clarify whether an economy has implemented the Model Law in full, in part, or based on its underlying principles. For example, note (a) indicates adoption except for the provisions on certification and electronic signatures.

[59] Adoption of a UNCITRAL Model Law in whole refers to national legislation that closely follows the full text or core provisions of the Model Law. Partial adoption means that only some provisions are incorporated, or the law is broadly influenced by the Model Law’s principles. The notes (a) to (c) on the UNCITRAL Model Law on Electronic Signatures status webpage clarify whether an economy has implemented the Model Law in full, in part, or based on its underlying principles. For example, note (a) indicates the legislation is influenced by the Model Law and the principles on which it is based.

Not in the UNCITRAL Model Law on Electronic Transferable Records

This indicator asks whether an economy has adopted the UNCITRAL Model Law on Electronic Transferable Records (MLETR) (2017). The MLETR is a model law on which an economy may base its legislation fully or in part. The MLETR ensures functionally equivalent to transferable documents or instruments. Transferable documents or instruments are paper-based documents or instruments that entitle the holder to claim an obligation and transfer that claim by handing over the document. Examples typically include bills of lading, bills of exchange, promissory notes and warehouse receipts (UNCITRAL, 2017) (box 6).

If the economy has neither adopted any parts of the mentioned international legal framework nor notified the adoption of the UNCITRAL Model Law on the UNCITRAL Secretariat, the score is '1'. If the economy has adopted fully or in part the Model Law,[60] but has not notified the UNCITRAL Secretariat, the score is '0.5'. If the economy has adopted fully or in part the Model Law and notified the UNICITRAL Model Law, the score assigned is '0'.

To see the status of the Model Law, check the UNCITRAL Model Law on Electronic Transferable Records (MLETR) (status) or the official government website.

Box 6

Relationship among the UNCITRAL Model Laws on Electronic Commerce, Electronic Signatures, and Electronic Transferable Records and the United Nations Convention on the Use of Electronic Communications

The MLEC is the first model legislative text that adopts the principles of non-discrimination, technological neutrality and functional equivalence. The principle of non-discrimination provides that a document must not be denied legal validity or enforceability solely on the grounds that it is in electronic form. The principle of technological neutrality mandates the adoption of neutral legislative provisions regarding the technology used. The functional equivalence principle recognizes that document and paper- and paper-based communications are given the same effect as their electronic counterparts. The MLES and MLETR apply the three principles established by the MLEC in recognizing the legal validity of electronic signatures and electronic transferable records.

The Electronic Communications Convention builds particularly on the MLEC and MLES and incorporates the principles of non-discrimination, technological neutrality, and functional equivalence. Certain provisions of the MLEC were amended by the Electronic Communications Convention in light of recent electronic commerce practices. As of 2025, 20 States were parties to the Convention.

The weights for each indicator

As shown in figure 29, the weights for each indicator are 22%, 11%, 11%, 7%, 7%, 7%, 7%, 7%,4%, 4%, 4%, 4%, and 4%. Maximum foreign equity shares for investment in the e-commerce sector, the first indicator, is given the greatest weight of 22% because limitations on foreign ownership discriminate and block foreign investment in the e-commerce sector.

[60] Adoption of a UNCITRAL Model Law in whole refers to national legislation that closely follows the full text or core provisions of the Model Law. Partial adoption means that only some provisions are incorporated, or the law is broadly influenced by the Model Law's principles. The single note on the UNCITRAL Model Law on Electronic Transferable Records status webpage clarifies whether an economy has implemented the Model Law in full, in part, or based on its underlying principles. Specifically, note (a) indicates the legislation is influenced by the Model Law and the principles on which it is based.

The second and the third indicators – requirements on online purchases and delivery, and licensing schemes for e-commerce providers – are equally assigned an equal weight of 11%. The requirements on sales *per se* or requirements on delivery set up direct barriers for vendors to engage in digital trade or reach consumers. Although e-commerce licensing improves reliability and confirms the existence of the business, the requirement is costly, and a rejection of licensing could preclude the e-commerce business from operating in certain economies.

The fourth indicator to the eighth indicator, which are online payment limitations, low *de minimis*, imposition of customs duties on electronic transmissions, domain name requirements and local presence requirements, are given a lesser weight of 7%. Each of them does not necessarily block sales or promotion of goods, but discourages e-commerce activities. The requirements for online payments could make cross-border paymentsin electronic commerce cumbersome. The low *de minimis* rule does not block flows of goodsbut tends to increase the prices of imports. The imposition of customs duties on electronic transmissions increases the costs of accessing digital services for consumers and businesses. Additionally, the domain name requirements, covering the commercial presence or a local presence requirement to purchase the local domain names, increase compliance costs. The use of the national domain also correlates with engagement of the locals and the businesses. Moreover, the local presence requirements for online service providershave an impact on businesses, especially SMEs.

Last, the ninth indicator to the last indicator is given the same weight of 4% since the lack ofa consumer protection law that applies to electronic commerce undermines trust in the eyes of consumers. The lack of legal recognition of electronic communications, transactions, and signatures creates uncertainty in the eyes of vendors. The international frameworks facilitate harmonized e-commerce regulations and practices; however, the absence of a legal framework or participation is not as impactful in terms of increasing the business compliance costs as the other indicators.